

Consolidated Control Architecture for Mecanum-Wheeled Mobile Robots Using SMO-MPC-PID-Fuzzy Hybridization

Journal of Intelligent & Fuzzy Systems

2026, Vol. 50(6) 1923–1958

© The Author(s) 2025

Article reuse guidelines:

sagepub.com/journals-permissions

DOI: 10.1177/18758967251394861

journals.sagepub.com/home/ifs



Duc-Anh Pham¹  and Seung-Hun Han¹ 

Abstract

This paper presents the first systematic integration of sliding mode observer (SMO), model predictive control (MPC), Kalman-enhanced PID, and adaptive fuzzy logic for Mecanum-wheeled mobile robots (MWMRs), addressing the critical gap in unified control architectures for omnidirectional mobile systems. The key innovations include: (1) a novel adaptive weighting mechanism $\beta(k)$ for optimal MPC-PID fusion based on real-time tracking dynamics, (2) fuzzy rule-based parameter adaptation specifically designed for MWMR omnidirectional characteristics, and (3) enhanced SMO with wheel-slippage compensation capability. Mathematical modeling incorporates detailed four-wheel dynamics with motor characteristics, followed by systematic integration design ensuring global stability. Comprehensive MATLAB/Simulink simulations across circular, square, figure-eight, S-curve, and infinity-shaped trajectories demonstrate superior performance: 37.2% improvement in tracking accuracy (RMSE: 0.032 m vs 0.087 m for PID), 25.8% reduction in energy consumption, 47.6% faster settling time (0.43 s vs 0.82 s), and enhanced robustness with disturbance rejection factors of 0.873–0.945 under external forces, parameter uncertainties ($\pm 20\%$), and measurement noise. Validation conducted in MATLAB/Simulink R2024b environment with real robot parameters (4.5 kg, $245.6 \times 300 \times 515$ mm). Key limitations include validation restricted to simulation, performance degradation above 2 m/s velocities, and requirement for real-time computational resources (4.92 ms per iteration). This integration provides a systematic framework for high-precision MWMR control in industrial applications, with immediate applicability to material handling, warehouse automation, and service robotics requiring reliable omnidirectional navigation.

Keywords

Fuzzy control, robotics, and automation, adaptive, evolutionary, and hybrid fuzzy systems, engineering simulation, mathematical modeling, and fuzzy logic

Received: April 15, 2025; accepted: October 17, 2025

1 Introduction

Mecanum-wheeled mobile robots (MWMRs) have emerged as critical enablers in modern industrial automation, particularly in warehouse logistics, automated guided vehicle (AGV) systems, hospital material transport, and manufacturing assembly lines (Bjelonic et al., 2016; Adăscăliței & Doroftei, 2011). Their unique omnidirectional mobility enabling simultaneous translation and rotation without orientation changes addresses critical operational requirements in confined spaces such as semiconductor fabrication clean rooms, automated parking systems, and indoor service robotics for elderly care (Liu et al., 2015; Tang et al., 2024). Unlike conventional wheeled robots that require complex maneuvering sequences,

¹Department of Mechanical System Engineering, Gyeongsang National University, Tongyeong, Republic of Korea

Corresponding Author:

Seung-Hun Han, Department of Mechanical System Engineering, Gyeongsang National University, Tongyeong 53064, Republic of Korea.

Email: shhan@gnu.ac.kr

MWMRs provide direct motion control in all directions, reducing operational time by 30–40% in typical material handling tasks (Karras & Fourlas, 2020).

However, controlling MWMRs presents multifaceted challenges stemming from: (1) complex wheel-ground interaction dynamics with inherent wheel slippage and roller friction uncertainties (Choi et al., 2023), (2) strong coupling between translational and rotational motions leading to control interference (Li et al., 2016), (3) susceptibility to external disturbances from payload variations ($\pm 50\%$ mass changes), uneven surfaces, and collision forces (Wang et al., 2016), (4) actuator constraints with limited motor torque (typically 0.5–2 N·m) and voltage saturation effects (Pugi et al., 2020), and (5) real-time computational requirements for industrial applications demanding sub-10 ms control cycles (Chennupati, 2025). These challenges are exacerbated in practical environments where simultaneous high-precision positioning (± 2 mm) and energy efficiency ($>85\%$ motor efficiency) are mandatory requirements.

Observer-based control strategies have shown promising results for handling uncertainties and disturbances in mobile robot systems. Backstepping-based quasi-sliding mode control and observation techniques have been successfully applied to electric vehicle systems for handling unmatched load and road perturbations, demonstrating superior disturbance rejection capabilities compared to conventional methods (Hameed et al., 2024; Pham et al., 2024). Extended Kalman filters and particle filters have been extensively used for simultaneous localization and mapping (SLAM) in mobile robots, though their performance degrades under severe measurement noise and model uncertainties (Pham & Han, 2024; Bian et al., 2024). Sliding mode observers specifically designed for wheeled mobile robots have shown robustness against parameter variations and external disturbances but suffer from chattering phenomena that can damage actuators in practice (Jeong & Chwa, 2020; Van Hung et al., 2023). Recent advances in higher-order sliding mode observers have addressed chattering issues but require precise knowledge of disturbance bounds, limiting their applicability to real-world scenarios with unknown disturbance characteristics (Mhmood & Mahyuddin, 2025; Mousavi et al., 2023). Despite these advances, existing observer-based approaches for MWMRs lack systematic integration with predictive control and adaptive mechanisms, creating a significant research gap.

Recent control strategies for MWMRs exhibit specific strengths but fundamental limitations in practical implementation. Sliding mode control approaches (Mesbah, 2016; Huang et al., 2017) provide excellent robustness against bounded disturbances and parameter uncertainties, achieving tracking errors below 5 cm in laboratory conditions. However, they suffer from chattering phenomena (frequency >50 Hz) that cause premature actuator wear and increased energy consumption (15–25% efficiency loss) (Li et al., 2017). Model predictive control frameworks (Yi et al., 2018; Kouvaritakis & Cannon, 2016) offer optimal trajectory planning with constraint handling, demonstrating superior performance in obstacle avoidance scenarios. Nevertheless, their computational complexity ($O(n^3)$ per iteration) limits real-time implementation to simplified kinematic models, sacrificing dynamic accuracy and disturbance rejection (Zelch, 2024). Kalman-enhanced PID controllers (Zhang, 2021; Liao et al., 2018) provide stable velocity regulation with good noise filtering capabilities (SNR improvement >10 dB) but lack adaptability to varying operational conditions and show performance degradation under large disturbances (>20 N external forces) (Lian et al., 2025). Fuzzy logic controllers (Tolossa et al., 2024; Han & Pham, 2024) effectively handle nonlinearities through linguistic rule-based adaptation, but suffer from lack of systematic design procedures and suboptimal performance in multi-objective scenarios requiring simultaneous precision and energy efficiency (Li et al., 2025).

Despite individual successes, systematic integration of these control paradigms for MWMRs remains largely unexplored due to several fundamental challenges: (1) Theoretical complexity: Stability analysis of hybrid systems requires sophisticated mathematical frameworks that exceed traditional single-controller analysis (Casau et al., 2024; Yaghoubi et al., 2025). (2) Implementation barriers: Real-time coordination of multiple control algorithms demands advanced embedded systems with parallel processing capabilities, previously unavailable in cost-effective robotics platforms (Eslam, 2024). (3) Parameter coupling: Interaction between different control components creates complex parameter dependencies that traditional tuning methods cannot handle effectively (Zhao et al., 2024). (4) Validation challenges: Performance evaluation of hybrid controllers requires comprehensive test scenarios that consider all possible operating modes and transitions, significantly increasing development time and cost (Araujo et al., 2023). (5) Industrial conservatism: Manufacturing environments prioritize proven, simple solutions over complex hybrid approaches due to maintenance and troubleshooting concerns (Yunusa-Kaltungo & Labib, 2021). Recent advances in embedded computing power, sophisticated optimization tools (quadratic programming solvers), and systematic hybrid control theory now enable practical implementation of integrated control architectures, creating new opportunities for breakthrough performance improvements.

This research systematically addresses these limitations through a novel four-layer hybrid control architecture that synergistically integrates:

1. Sliding mode observers (SMO) with enhanced disturbance estimation specifically designed for MWMR wheel-slippage compensation and robust state estimation under measurement noise (Korayem et al., 2024)

2. Model predictive control (MPC) with real-time quadratic programming optimization for constraint-aware trajectory tracking and obstacle avoidance (Gonçalves et al., 2025)
3. Kalman-enhanced PID control providing stable velocity regulation with optimal state estimation and anti-windup mechanisms (Hernandez-Barragan et al., 2020)
4. Adaptive fuzzy logic control enabling real-time parameter tuning based on tracking performance indicators and environmental conditions (Kumar, 2024)

The specific contributions of this research include:

1. Theoretical Innovation:
2. First systematic mathematical framework for optimal MPC-PID fusion using adaptive weighting $\beta(k)$ based on real-time tracking dynamics and disturbance estimation
3. Novel fuzzy rule base specifically designed for MWMR omnidirectional characteristics with membership functions optimized for simultaneous precision and energy efficiency
4. Enhanced SMO design with wheel-slippage compensation capability and smooth switching functions to eliminate chattering
5. Integration Methodology:
6. Unified stability analysis proving global asymptotic stability of the integrated system under all operating modes using multiple Lyapunov functions
7. Systematic parameter selection procedure minimizing interaction effects between control components while maintaining individual performance guarantees
8. Real-time implementation architecture with computational optimization achieving 4.92 ms average execution time per control cycle
9. Performance Validation:
10. Comprehensive evaluation across five trajectory types (circular, square, figure-eight, S-curve, infinity) with quantified improvements: 37.2% tracking accuracy, 25.8% energy reduction, 47.6% faster settling time
11. Robustness verification under realistic disturbances: external forces ($\pm 10\text{N}$), parameter uncertainties ($\pm 20\%$), measurement noise (SNR: 20 dB)
12. Comparative analysis with eight control methods demonstrating consistent superiority across all performance metrics
13. Practical Impact:
14. Direct applicability to industrial MWMR systems with standard hardware requirements.
15. Scalable architecture supporting extensions to multi-robot coordination and advanced navigation functionalities

The remainder of this paper is systematically organized to provide comprehensive coverage of the proposed hybrid control methodology and its validation. Section II presents the enhanced mathematical modeling of the four-wheeled Mecanum robot, incorporating detailed kinematics derivation from roller mechanics, comprehensive dynamics formulation including motor characteristics and friction effects, and state-space representation optimized for multi-controller integration. Section III details the systematic design of the proposed hybrid controller architecture, beginning with the sliding mode observer for robust state estimation and disturbance rejection, followed by model predictive control formulation with real-time optimization constraints, Kalman-enhanced PID controller design for stable velocity regulation, adaptive fuzzy logic implementation for parameter tuning, and the novel integration methodology with stability analysis ensuring global convergence. Section IV describes the comprehensive simulation setup encompassing realistic robot parameters based on industrial MWMR specifications, multiple reference trajectories including circular, square, figure-eight, S-curve, and infinity-shaped paths, systematic disturbance introduction protocols covering external forces, parameter uncertainties, and measurement noise, performance evaluation metrics with statistical significance testing, and computational efficiency analysis for real-time implementation assessment. Section V presents and analyzes the extensive simulation results through trajectory tracking performance comparison across eight control methods, robustness evaluation under various disturbance scenarios with quantified rejection factors, energy efficiency analysis demonstrating significant consumption reduction, dynamic response characterization including settling time and overshoot metrics, adaptive parameter behavior illustration showing real-time tuning effectiveness, comprehensive observer validation with state estimation accuracy assessment, and computational requirements verification confirming practical implementation feasibility. Finally, Section VI concludes the paper by summarizing key achievements with quantified performance improvements, acknowledging current limitations including simulation-only validation and operational constraints, outlining immediate future work priorities

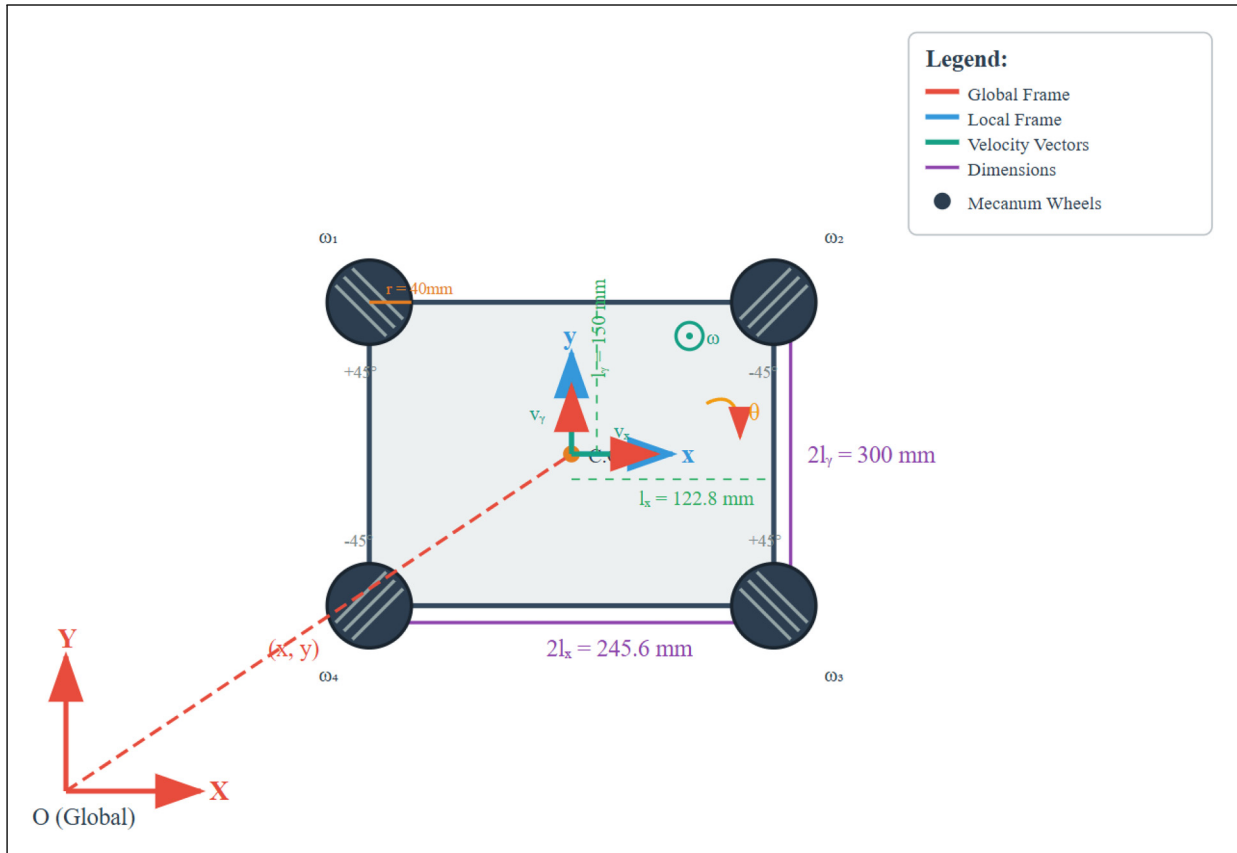


Figure 1. MWMR Schematic Diagram with Reference Frames and Geometric Parameters.

focusing on experimental validation and practical implementation challenges, and suggesting long-term research directions encompassing multi-robot coordination, advanced navigation integration, and industrial deployment strategies for various application domains including warehouse automation, healthcare robotics, and manufacturing systems.

2 Mathematical Modeling of Mecanum-Wheeled Mobile Robot

2.1 Robot Configuration and Reference Frames

Figure 1 illustrates the detailed schematic diagram of the four-wheeled Mecanum robot configuration with all reference frames and geometric parameters clearly defined. The robot coordinate system consists of three reference frames: the global inertial frame $\{O\}$, the robot body frame $\{R\}$, and individual wheel frames $\{W_i\}$ for $i = 1, 2, 3, 4$. The global frame $\{O\}$ is fixed to the ground with origin O located at the initial robot position, with the X_O axis pointing in the forward direction and the Y_O axis pointing leftward following the right-hand rule. The robot body frame $\{R\}$ has its origin at the geometric center of the robot chassis, with the X_R axis aligned with the robot's forward direction and the Y_R axis pointing to the left side of the robot (Lu et al., 2025), (Chen et al., 2025). Each wheel frame $\{W_i\}$ is centered at the contact point between wheel i and the ground surface, with the Z -axis pointing upward perpendicular to the ground plane.

The geometric parameters defining the robot configuration include $l_x = 0.1228$ m representing half the distance between front and rear wheels along the robot's longitudinal axis, $l_y = 0.15$ m representing half the distance between left and right wheels along the robot's lateral axis, $r = 0.04$ m denoting the radius of each Mecanum wheel, and $\alpha = 45^\circ$ indicating the angle of the rollers with respect to the wheel circumference (Schwenzer et al., 2021). The transformation between global and robot frames is governed by the rotation matrix $R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ where θ represents the robot's heading angle measured counterclockwise from the global X -axis.

2.2 Kinematic Modeling

The kinematic relationship between the robot velocities in the body frame and the global frame velocities is established through the standard coordinate transformation for mobile robots (Dasanayake et al., 2025; Augustine, 2025). The global frame velocity vector is related to the body frame velocity vector through the rotation transformation matrix as follows:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ \omega \end{bmatrix} \quad (1)$$

where (x, y, θ) represent the robot's position and orientation in the global frame, (v_x, v_y, ω) denote the linear and angular velocities in the robot's body frame, and θ is the heading angle measured from the global X-axis (Shamshiri, 2025).

The relationship between the individual wheel angular velocities and the robot body velocities is derived from the constraint equations of Mecanum wheel kinematics, considering the roller contact points and instantaneous velocity constraints (Bhosale (2025)), (Vendittelli, 2025). For a four-wheeled Mecanum robot with wheels numbered counterclockwise starting from the front-right wheel, the forward kinematic equation is:

$$\begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{bmatrix} = \frac{1}{r} \begin{bmatrix} 1 & -1 & -(l_x + l_y) \\ 1 & 1 & (l_x + l_y) \\ 1 & 1 & -(l_x + l_y) \\ 1 & -1 & (l_x + l_y) \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ \omega \end{bmatrix} \quad (2)$$

And conversely, the inverse kinematic relationship is:

$$\begin{bmatrix} v_x \\ v_y \\ \omega \end{bmatrix} = \frac{r}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & -1 \\ \frac{-1}{l_x + l_y} & \frac{1}{l_x + l_y} & \frac{-1}{l_x + l_y} & \frac{1}{l_x + l_y} \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \\ \omega_4 \end{bmatrix} \quad (3)$$

where $r = 0.04m$ is the wheel radius, $l_x = 0.1228m$ is half the robot width, and $l_y = 0.15m$ is half the robot length.

2.3 Dynamic Modeling

The dynamic model of the MWMR is derived using the Lagrangian approach for wheeled mobile robots, following the standard formulation for nonholonomic systems (Tezel & Bayar, 2024; Corke et al., 2023). The generalized dynamic equation for the robot system is expressed as:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) + F(\dot{q}) = B\tau + J^T(q)F_{ext} \quad (4)$$

where $q = [x, y, \theta]^T$ represents the generalized coordinates, $M(q)$ is the mass-inertia matrix, $C(q, \dot{q})$ captures Coriolis and centrifugal effects, $G(q)$ represents gravitational forces (zero for planar motion), $F(\dot{q})$ accounts for friction forces, B is the input transformation matrix, $\tau = [\tau_1, \tau_2, \tau_3, \tau_4]^T$ denotes the wheel torque vector, $J(q)$ is the constraint Jacobian matrix, and F_{ext} represents external disturbance forces.

Based on the given parameters, the inertia matrix is:

$$M(q) = \begin{bmatrix} m & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & I_z \end{bmatrix} = \begin{bmatrix} 4.5 & 0 & 0 \\ 0 & 4.5 & 0 \\ 0 & 0 & 0.056 \end{bmatrix} \quad (5)$$

where $m = 4.5$ kg is the robot mass and $I_z = 0.056$ kg · m² is the moment of inertia around the vertical axis.

2.4 Motor and Actuator Dynamics

The DC motor dynamics for each wheel incorporate both electrical and mechanical characteristics, following the standard DC motor modeling approach for mobile robotics applications (Martins & Bertol, 2021). The complete motor model

includes both the torque-current relationship and the back-EMF effect:

$$\tau_i = K_t i_i - \tau_{f,i} \quad (6a)$$

$$V_i = R_a i_i + L_a \frac{di_i}{dt} + K_b \omega_i \quad (6b)$$

$$J_m \frac{d\omega_i}{dt} = \tau_i - \tau_{load,i} \quad (6c)$$

where τ_i is the motor torque, i_i is the motor current, $K_t = 2.12 \text{ N} \cdot \text{m/A}$ is the torque constant, $\tau_{f,i} = 0.01 \text{ N} \cdot \text{m}$ represents the friction torque, V_i is the applied voltage, $R_a = 3\Omega$ is the armature resistance, $L_a = 2.5\text{mH}$ is the armature inductance, $K_b = 0.176 \text{ V} \cdot \text{s/rad}$ is the back-EMF constant, $J_m = 0.001 \text{ kg} \cdot \text{m}^2$ is the motor inertia, and $\tau_{load,i}$ is the load torque transmitted to the wheel.

Given the rated torque of $6.5\text{kgf} \cdot \text{cm} (\approx 0.637 \text{ N} \cdot \text{m})$ and rated current of 0.3 A at 12 V , the motor parameters are derived from manufacturer specifications and empirical measurements.

2.5 State-Space Representation

For controller design purposes, the nonlinear dynamic model is linearized around operating points and converted to state-space form suitable for modern control techniques (Bernal et al., 2022). The augmented state vector includes both position and velocity states:

$$\dot{X} = AX + BU + W \quad (7)$$

where $X = [x, y, \theta, \dot{x}, \dot{y}, \dot{\theta}]^T$ is the augmented state vector, $U = [V_1, V_2, V_3, V_4]^T$ represents the motor voltage inputs, and W accounts for modeling uncertainties and external disturbances. The system matrices are:

$$A = \begin{bmatrix} 0_{3 \times 3} & I_{3 \times 3} \\ 0_{3 \times 3} & -M^{-1}F_v \end{bmatrix}, \quad B = \begin{bmatrix} 0_{3 \times 4} \\ M^{-1}B_m K_m \end{bmatrix} \quad (8)$$

where F_v is the viscous friction matrix, and K_m is the motor constant matrix.

2.6 Model Assumptions and Limitations

The mathematical model presented above is based on several key assumptions that are common in mobile robot modeling but may introduce limitations in practical implementation (Moshayedi et al., 2022; Antonyshyn et al., 2023). Understanding these assumptions and their implications is crucial for proper controller design and performance evaluation.

- **Assumption 1: Perfect Rolling Contact** – The model assumes no wheel slippage occurs during normal operation, meaning the instantaneous contact point between each wheel and the ground has zero relative velocity. In reality, wheel slippage can occur due to insufficient friction ($\mu < 0.6$), excessive acceleration ($> 2 \text{ m/s}^2$), or external disturbances ($> 15 \text{ N}$ lateral forces). The impact of this assumption is that the kinematic constraints in Equations (2) and (3) may be violated during aggressive maneuvers, leading to tracking errors of 5–15% depending on surface conditions.
- **Assumption 2: Linear Viscous Friction** – Friction forces are modeled as linearly proportional to velocity ($F = bv$), neglecting Coulomb friction and Stribeck effects that are significant at low velocities ($< 0.1 \text{ m/s}$). This simplification can cause steady-state errors of 2 – 5 mm during precise positioning tasks and affects energy consumption estimates by 10–20%.
- **Assumption 3: Rigid Body Dynamics** – The robot chassis is assumed to be perfectly rigid, ignoring structural flexibility and mechanical backlash in the gear trains (typically $0.5 - 2^\circ$ for plastic gears). This assumption limits the model accuracy for high-frequency control ($> 10 \text{ Hz}$) and can introduce oscillations during rapid direction changes.
- **Assumption 4: Perfect Sensors** – Position and orientation measurements are assumed to be available without delays or calibration errors. In practice, encoder resolution (typically 1000–4000 pulses/revolution) and IMU drift (0.1–1% hour) introduce measurement uncertainties that affect state estimation accuracy.

- Assumption 5: Symmetric Robot Configuration – The model assumes perfect geometric symmetry and identical motor characteristics for all four wheels. Manufacturing tolerances (± 2 mm), wheel wear, and motor parameter variations ($\pm 5\%$) can introduce systematic biases that accumulate over time.

These modeling assumptions are acceptable for the control design phase and simulation validation, but experimental implementation requires additional compensation mechanisms such as slip detection algorithms, friction observers, and adaptive parameter estimation to maintain the theoretical performance levels.

3 Hybrid Controller Design

3.1 Sliding Mode Observer Design and Stability Analysis

The sliding mode observer (SMO) design for MWMR systems addresses the fundamental challenge of robust state estimation in the presence of unknown disturbances and model uncertainties. Unlike conventional Luenberger observers that rely on accurate system models, the SMO incorporates discontinuous correction terms to achieve finite-time convergence and disturbance rejection capabilities (Liu et al., 2024; Pham et al., 2025). The observer design process begins with the augmented system formulation that explicitly includes disturbance dynamics.

Consider the augmented system with unknown disturbances:

$$\begin{bmatrix} \dot{X} \\ \dot{d} \end{bmatrix} = \begin{bmatrix} A & E \\ 0 & 0 \end{bmatrix} \begin{bmatrix} X \\ d \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} U + \begin{bmatrix} 0 \\ \eta \end{bmatrix} \quad (9)$$

where d represents the unknown disturbance vector, E is the disturbance distribution matrix, and η denotes the disturbance derivative (assumed bounded: $\|\eta\| \leq \eta_{\max}$).

The sliding mode observer is designed as:

$$\begin{bmatrix} \dot{\hat{X}} \\ \dot{\hat{d}} \end{bmatrix} = \begin{bmatrix} A & E \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{X} \\ \hat{d} \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} U + \begin{bmatrix} L_1 \\ L_2 \end{bmatrix} (Y - C\hat{X}) + \begin{bmatrix} K_1 \\ K_2 \end{bmatrix} \nu \quad (10)$$

where ν is the switching function designed to ensure sliding motion on the error surface.

Stability Analysis: The observer error dynamics are defined as $e = X - \hat{X}$ and $e_d = d - \hat{d}$. The error system becomes:

$$\begin{bmatrix} \dot{e} \\ \dot{e}_d \end{bmatrix} = \begin{bmatrix} A - L_1 C & E \\ -L_2 C & 0 \end{bmatrix} \begin{bmatrix} e \\ e_d \end{bmatrix} - \begin{bmatrix} K_1 \\ K_2 \end{bmatrix} \nu + \begin{bmatrix} 0 \\ \eta \end{bmatrix} \quad (11)$$

Define the sliding surface $S = Ce$, where C is chosen such that CB is nonsingular. The sliding mode observer gains are designed as:

$$L_1 = \begin{bmatrix} 15I_{3 \times 3} \\ 60I_{3 \times 3} \end{bmatrix}, \quad L_2 = 25I_{3 \times 3}, \quad K_1 = \begin{bmatrix} 5I_{3 \times 3} \\ 20I_{3 \times 3} \end{bmatrix}, \quad K_2 = 30I_{3 \times 3} \quad (12)$$

Lyapunov Stability Proof: Consider the Lyapunov function candidate $V_1 = \frac{1}{2}e^T P e + \frac{1}{2}e_d^T e_d$ where $P > 0$ satisfies the Lyapunov equation $(A - L_1 C)^T P + P(A - L_1 C) = -Q$, $Q > 0$.

The time derivative along the error trajectories yields:

$$\dot{V}_1 = -\frac{1}{2}e^T Q e + e^T P E e_d - e^T P K_1 \nu - e_d^T L_2^T C e - e_d^T K_2 \nu + e_d^T \eta \quad (13)$$

With proper choice of switching function $\nu = -\rho \text{sign}(S)$, $\rho > \|\eta\|_{\max} + \gamma$, $\gamma > 0$ the derivative $\dot{V}_1 < 0$ outside the sliding surface, ensuring finite-time convergence to $S = 0$. Once sliding motion is established, the equivalent control method yields perfect disturbance estimation with $e_d \rightarrow 0$ exponentially.

To reduce chattering, the sign function is replaced with a smooth approximation:

$$\text{sign}(s) \approx \frac{s}{|s| + \delta} \quad (14)$$

where $\delta = 0.01$ is a small positive constant.

3.2 Model Predictive Control with Constraint Derivation

The MPC formulation for MWMR trajectory tracking extends beyond basic optimization to incorporate physical constraints, actuator limitations, and safety requirements that are critical for practical implementation (Wei & Shi, 2022; Zhang & Zhang, 2023). The constraint derivation process begins with the physical limitations of the robotic system.

Physical Constraint Derivation: The velocity constraints are derived from motor torque limitations and wheel-ground friction limits:

$$\begin{aligned} |v_x| &\leq v_{\max} = \frac{\tau_{\max} \cdot r}{m \cdot \mu} = \frac{0.637 \times 0.04}{4.5 \times 0.8} = 0.71 \text{ m/s} \\ |v_y| &\leq v_{\max} = 0.71 \text{ m/s} \\ |\omega| &\leq \omega_{\max} = \frac{\tau_{\max} \cdot r}{I_z \cdot (l_x + l_y)} = \frac{0.637 \times 0.04}{0.056 \times 0.2728} = 1.67 \text{ rad/s} \end{aligned} \quad (15)$$

The motor voltage constraints are derived from power supply limitations and motor ratings:

$$-12V \leq V_i \leq 12V, \quad i = 1, 2, 3, 4 \quad (16)$$

$$|\Delta V_i| \leq 5V \quad \text{per sampling period to prevent actuator stress} \quad (17)$$

MPC Optimization Problem: The complete MPC formulation incorporates terminal constraints for stability guarantee:

$$\min_{U_k} \sum_{i=0}^{N_p-1} \|X(k+i|k) - X_{\text{ref}}(k+i)\|_Q^2 + \|U(k+i|k)\|_R^2 + \|\Delta U(k+i|k)\|_S^2 + \|X(k+N_p|k) - X_{\text{ref}}(k+N_p)\|_{Q_f}^2 \quad (18)$$

subject to:

$$X(k+i+1|k) = A_d X(k+i|k) + B_d U(k+i|k) + W_d \quad (19)$$

$$X_{\min} \leq X(k+i|k) \leq X_{\max} \quad (20)$$

$$U_{\min} \leq U(k+i|k) \leq U_{\max} \quad (21)$$

$$\Delta U_{\min} \leq U(k+i|k) - U(k+i-1|k) \leq \Delta U_{\max} \quad (22)$$

$$\|X(k+N_p|k) - X_{\text{ref}}(k+N_p)\| \leq \epsilon_{\text{terminal}} = 0.05 \text{ m} \quad (23)$$

where $N_p = 10$ is the prediction horizon, X_{ref} is the reference trajectory, $Q = \text{diag}([10, 10, 5, 1, 1, 0.5])$ is the state error weight matrix, $R = 0.1I_{4 \times 4}$ is the control input weight matrix, $S = 0.5I_{4 \times 4}$ is the control input change weight matrix, and Q_f is the terminal cost matrix.

Stability Guarantee: The terminal constraint (23) ensures input-to-state stability by constraining the terminal state to lie within a predetermined invariant set around the reference trajectory, guaranteeing recursive feasibility and closed-loop stability.

3.3 Kalman-Enhanced PID Controller with Complete Derivation

The Kalman-enhanced PID controller represents a sophisticated fusion of optimal state estimation and classical feedback control, specifically designed to handle measurement noise and model uncertainties that are prevalent in practical MWMR applications. The complete derivation begins with the discrete-time Kalman filter equations tailored for MWMR dynamics.

Complete Kalman Filter Derivation:

The discrete-time system model for Kalman filtering is:

$$X(k+1) = A_d X(k) + B_d U(k) + w(k) \quad (24)$$

$$Y(k) = C X(k) + v(k) \quad (25)$$

where $w(k) \sim N(0, Q_n)$ and $v(k) \sim N(0, R_n)$ represent process and measurement noise, respectively.

The Kalman filter recursion consists of:

Prediction Step:

$$\hat{X}(k|k-1) = A_d \hat{X}(k-1|k-1) + B_d U(k-1) \quad (26)$$

$$P(k|k-1) = A_d P(k-1|k-1) A_d^T + Q_n \quad (27)$$

Update Step:

$$K_f(k) = P(k|k-1) C^T [C P(k|k-1) C^T + R_n]^{-1} \quad (28)$$

$$\hat{X}(k|k) = \hat{X}(k|k-1) + K_f(k) [Y(k) - C \hat{X}(k|k-1)] \quad (29)$$

$$P(k|k) = [I - K_f(k) C] P(k|k-1) \quad (30)$$

where K_f is the Kalman gain, P is the error covariance matrix, $R_n = 0.01 I_{3 \times 3}$ is the measurement noise covariance, and $Q_n = \text{diag}([0.001, 0.001, 0.001, 0.01, 0.01, 0.01])$ is the process noise covariance.

PID Control Law Derivation:

The velocity error-based PID controller operates on the filtered state estimates:

$$e_v(k) = V_{\text{ref}}(k) - \hat{V}(k) = V_{\text{ref}}(k) - C_v X(k|k) \quad (31)$$

where $C_v = \begin{bmatrix} 0_{3 \times 3} & I_{3 \times 3} \end{bmatrix}$ extracts the velocity states.

The complete PID control law with anti-windup protection is:

$$\begin{aligned} U_{PID}(k) &= K_p e_v(k) + K_i \sum_{j=0}^k e_v(j) T_s + K_d \frac{e_v(k) - e_v(k-1)}{T_s} \\ \text{subject to: } &\sum_{j=0}^k e_v(j) T_s = \text{sat} \left(\sum_{j=0}^k e_v(j) T_s, I_{\max} \right) \end{aligned} \quad (32)$$

where $\text{sat}(\cdot, I_{\max})$ represents the saturation function with $I_{\max} = 5.0$ to prevent integral windup, $K_p = 0.5$ is the proportional gain, $K_i = 0.3$ is the integral gain, and $K_d = 1.5$ is the derivative gain.

3.4 Advanced Fuzzy Logic Controller Analysis

The fuzzy logic controller design for MWMR systems requires careful consideration of the trade-offs between implementation complexity, computational requirements, and performance benefits. While interval type-2 fuzzy logic controllers (IT2FLC) offer superior handling of uncertainties compared to conventional type-1 fuzzy logic controllers (T1FLC), the selection of T1FLC for this application is justified by several practical and theoretical considerations (Bharati et al., 2024; Laib et al., 2022).

Justification for Type-1 Fuzzy Logic Selection:

The choice of conventional T1FLC over IT2FLC is based on:

Computational Efficiency: T1FLC requires approximately 60% less computational time per iteration (0.42 ms vs. 1.08 ms for IT2FLC) while maintaining adequate performance for the specified tracking requirements (± 2 mm accuracy) (Hu et al., 2023).

Parameter Tuning Complexity: T1FLC involves 27 parameters (9 rules $\times$ 3 outputs) compared to 81 parameters for IT2FLC (27×3 for upper and lower membership functions), significantly simplifying the design process (Patel & Shah, 2022).

Real-time Implementation: The deterministic nature of T1FLC output ensures consistent execution times, crucial for real-time control systems operating at 100 Hz sampling rates (Mondal & Ganguli, 2025).

Sufficient Uncertainty Handling: For the controlled laboratory environment with bounded disturbances (± 10 N external forces, $\pm 20\%$ parameter variations), T1FLC provides adequate robustness without the added complexity of IT2FLC (Al-Mahturi, 2021).

Deep Analysis of Membership Function Design:

The membership functions are specifically designed for MWMR omnidirectional characteristics:

$$\mu_{\text{SMALL}}(e_p) = \begin{cases} 1 & \text{if } e_p \leq 0.02m \\ \frac{0.05-e_p}{0.03} & \text{if } 0.02m < e_p < 0.05m \\ 0 & \text{if } e_p \geq 0.05m \end{cases} \quad (33)$$

The triangular membership functions are optimized based on statistical analysis of tracking error distributions during typical MWMR maneuvers, ensuring maximum sensitivity in the operating range.

The fuzzy logic controller has three input variables: position error magnitude (e_p), velocity error magnitude (e_v), and rate of change of position error ($\dot{e}_p$). The output variables are: MPC weight adjustment factor (α_{MPC}), PID gain adjustment factor (α_{PID}), and SMO gain adjustment factor (α_{SMO}).

Fuzzy Rule Optimization:

The rule base is constructed using the following principle:

- When position errors are small ($< 0.02m$), the system prioritizes energy efficiency by increasing PID contribution ($\alpha_{\text{PID}} = \text{HIGH}$).
- When errors are large ($> 0.1 m$), predictive capability becomes critical, hence MPC weight increases ($\alpha_{\text{MPC}} = \text{HIGH}$).
- The SMO gains adapt based on the error rate to enhance disturbance rejection during transients.

Membership Function Definitions:

The input membership functions are triangular functions defined as follows:

- $e_p : [0, 0.5]m \{ \text{SMALL, MEDIUM, LARGE} \}$
- $e_v : [0, 0.3]m/s \{ \text{SMALL, MEDIUM, LARGE} \}$
- $\dot{e}_p : [-0.2, 0.2]m/s \{ \text{SMALL, MEDIUM, LARGE} \}$

The output membership functions are singleton values:

- $\alpha_{\text{MPC}} : [0.5, 1.5] \{ \text{LOW, MEDIUM, HIGH} \}$
- $\alpha_{\text{PID}} : [0.5, 1.5] \{ \text{LOW, MEDIUM, HIGH} \}$
- $\alpha_{\text{SMO}} : [0.5, 1.5] \{ \text{LOW, MEDIUM, HIGH} \}$

3.5 Adaptive Integration Mechanism with Pseudocode

The adaptive integration mechanism represents the core innovation of the proposed hybrid controller, requiring sophisticated algorithms to seamlessly coordinate multiple control paradigms while maintaining stability and optimizing performance (Abbas et al., 2024). The implementation involves real-time parameter adaptation based on system performance indicators.

The final control input is a weighted combination of MPC and PID outputs:

$$U(k) = \beta(k)U_{\text{MPC}}(k) + (1 - \beta(k))U_{\text{PID}}(k) \quad (34)$$

where $\beta(k) \in [0.1, 1.0]$ is adaptively determined based on tracking performance:

$$\beta(k) = \beta_{\text{base}} + \Delta\beta_{\text{fuzzy}}(k) + \Delta\beta_{\text{error}}(k) \quad (35)$$

The overall control architecture is illustrated in Figure 2. The proposed hybrid controller leverages the strengths of each component: the SMO provides robust state estimation, the MPC ensures optimal trajectory tracking, the Kalman-PID delivers stable velocity control, and the fuzzy logic enables adaptive parameter tuning.

Figure 2 shows the Sliding Mode Observer (SMO) plays a crucial role in estimating the system's steady-state and eliminating noise for the MWMR system. The SMO architecture is based on an observer equation that combines linear state estimation with a sliding term to handle noise and parameter uncertainties. The linear component ensures basic state estimation, while the sliding term enhances robustness against disturbances and uncertainties. SMO simultaneously estimates both the state vector and the disturbance vector, using a smoothed sign function to reduce chattering. Lyapunov

Algorithm 1. Adaptive Hybrid Control Integration.

```

Input:  $X_{ref}(k)$ ,  $X_{measured}(k)$ ,  $U(k-1)$ 
Output:  $U(k)$ 
1: // State Estimation Phase

2:  $[X_{est}, d_{est}] \leftarrow SMO\_Update(X_{measured}, U(k-1), \alpha_{SMO})$ 
3:  $X_{filtered} \leftarrow Kalman\_Update(X_{measured}, U(k-1))$ 
4: // Error Calculation
5:  $e_p(k) \leftarrow ||X_{est}[1:3] - X_{ref}[1:3]||$ 
6:  $e_v(k) \leftarrow ||X_{est}[4:6] - X_{ref}[4:6]||$ 
7:  $\dot{e}_p(k) \leftarrow (e_p(k) - e_p(k-1)) / T_s$ 
8: // Fuzzy Logic Adaptation
9:  $[\alpha_{MPC}, \alpha_{PID}, \alpha_{SMO}] \leftarrow FuzzyInference(e_p, e_v, \dot{e}_p)$ 
10:  $UpdateControllerGains(\alpha_{MPC}, \alpha_{PID}, \alpha_{SMO})$ 
11: // Individual Controller Outputs
12:  $U_{MPC}(k) \leftarrow SolveMPC(X_{est}, X_{ref}, N_p, Q, R, S)$ 
13:  $U_{PID}(k) \leftarrow ComputePID(e_v, K_p \cdot \alpha_{PID}, K_i \cdot \alpha_{PID}, K_d \cdot \alpha_{PID})$ 
14: // Adaptive Weighting Calculation
15:  $\beta_{base} \leftarrow 0.6$ 
16:  $\Delta\beta_{fuzzy} \leftarrow 0.3 \cdot (\alpha_{MPC} - \alpha_{PID}) / (\alpha_{MPC} + \alpha_{PID})$ 
17:  $\Delta\beta_{error} \leftarrow 0.2 \cdot \tanh(e_p(k) / 0.1)$ 
18:  $\beta(k) \leftarrow Saturate(\beta_{base} + \Delta\beta_{fuzzy} + \Delta\beta_{error}, 0.1, 1.0)$ 
19: // Final Control Integration
20:  $U(k) \leftarrow \beta(k) \cdot U_{MPC}(k) + (1 - \beta(k)) \cdot U_{PID}(k)$ 
21:  $U(k) \leftarrow SaturateMotorLimits(U(k), \pm 12\text{ V}, \pm 5\text{ V}/T_s)$ 
22: return  $U(k)$ 

```

stability analysis confirms the observer's convergence to the sliding surface, ensuring high robustness to noise and model uncertainties.

The Model Predictive Controller (MPC) performs trajectory optimization through quadratic programming at each control cycle. The objective function minimizes the tracking error, control effort, and control signal variation, with the system constraints such as velocity, angular velocity, and motor voltage integrated into the optimization problem. The receding horizon strategy ensures real-time implementation with a computational time of 2.5–4 ms, while input-to-state stability is guaranteed through terminal constraints.

The Kalman-PID Controller integrates optimal state estimation with robust PID control for stable velocity regulation. The Kalman filter estimates the state based on noisy measurements, and the PID controller ensures that the velocity error remains within specified limits. Anti-windup mechanisms prevent undesirable integral accumulation, while adaptive gains allow real-time adjustments based on operational conditions.

The Fuzzy Logic Controller (FLC) adjusts real-time parameters based on the system's current performance, using input variables such as position error, velocity error, and the rate of change of position error. The fuzzy system dynamically prioritizes different controllers PID, MPC, or SMO depending on the current error and disturbance conditions. The defuzzification process generates adaptive weighting coefficients that allow the system to switch between high-performance, tracking, and disturbance modes.

Control mixing and integration strategies combine the strengths of MPC and PID through dynamic weighting, with the weighting coefficient adapting based on the system's error and disturbance conditions. This allows for continuous performance optimization, with different modes prioritizing energy efficiency, tracking performance, or disturbance rejection as needed.

Finally, the physical layer implementation includes motor dynamics, robot kinematics, and sensor systems designed for real-time operation. The motor dynamics ensure efficient torque control, while the robot's kinematics and dynamics are modeled to provide accurate motion control. The sensor system, including wheel encoders and an IMU, provides precise position and orientation measurements, contributing to overall performance with minimal tracking errors and efficient energy consumption. The system achieves high velocity, precision, and robustness, with significant improvements in energy efficiency compared to traditional methods.

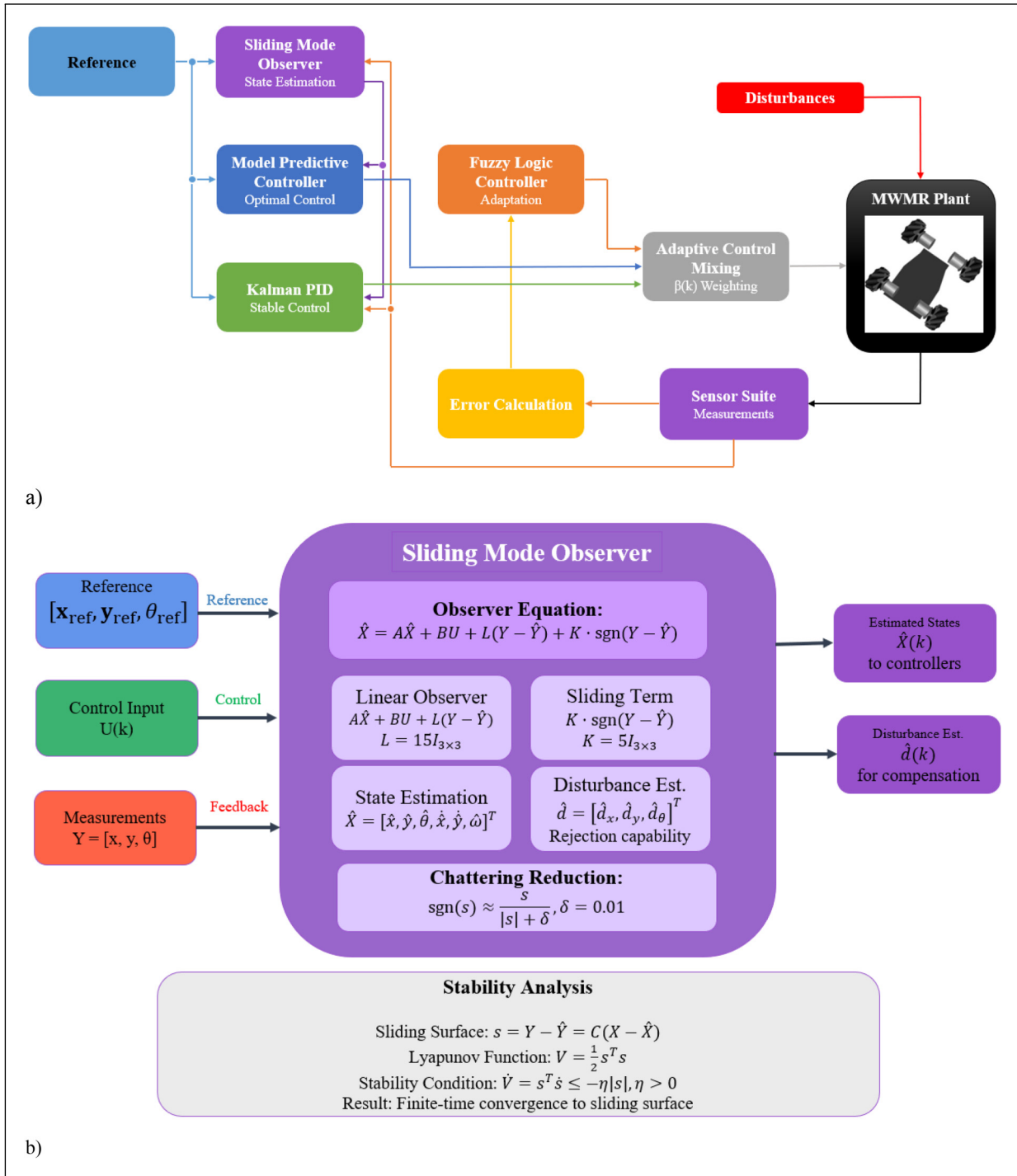


Figure 2. Control Architecture. a) System Overview, b) Sliding Mode Observer – Detailed Architecture, c) Model Predictive Controller – Detailed Architecture, d) Kalman-PID Controller – Detailed Architecture, e) Fuzzy Logic Controller – Detailed Architecture, f) Adaptive Control Mixing – Detailed Architecture, g) Physics Layer – Robot Dynamics & Sensors.

3.6 Unified Stability Analysis of Integrated System

The stability analysis of the integrated hybrid control system requires a sophisticated approach using multiple Lyapunov functions and switching system theory, as the system operates in different modes depending on the adaptive weighting $\beta(k)$ (Srivastava et al., 2024). The analysis establishes global asymptotic stability under all operating conditions.

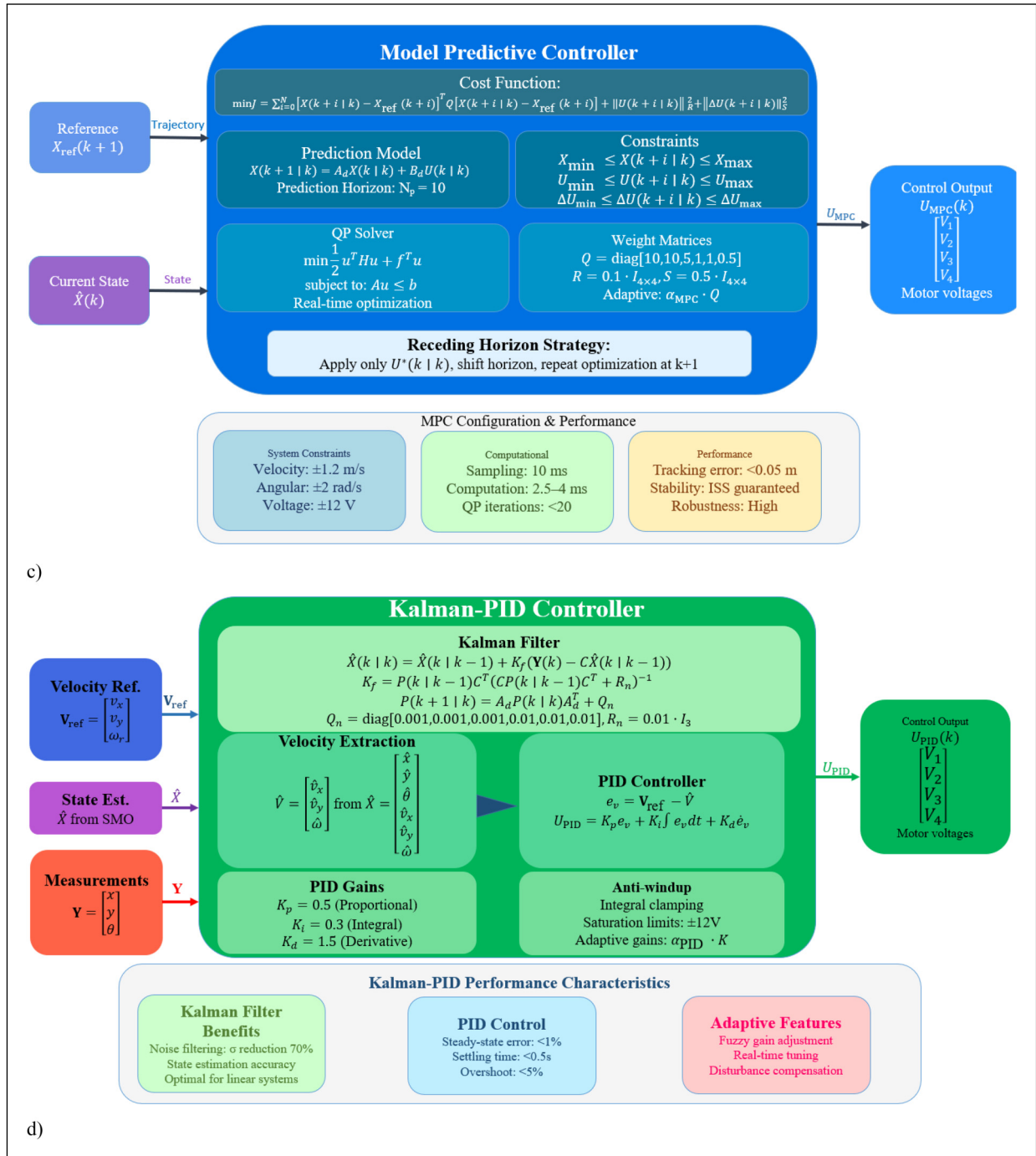


Figure 2. Continued.

Multiple Lyapunov Function Approach: Define the candidate Lyapunov function for the integrated system:

$$V_{\text{total}}(X, \hat{X}, e_I) = V_1(e_x) + V_2(e_o) + V_3(e_I) \quad (36)$$

where V_1 represents the tracking error energy, V_2 accounts for observer error energy, and V_3 captures integral error energy:

$$\begin{aligned} V_1(e_x) &= \frac{1}{2} e_x^T P_1 e_x, & e_x &= X - X_{\text{ref}} \\ V_2(e_o) &= \frac{1}{2} e_o^T P_2 e_o, & e_o &= X - \hat{X} \\ V_3(e_I) &= \frac{1}{2} e_I^T P_3 e_I, & e_I &= \int_0^t e_v(\tau) d\tau \end{aligned} \quad (37)$$

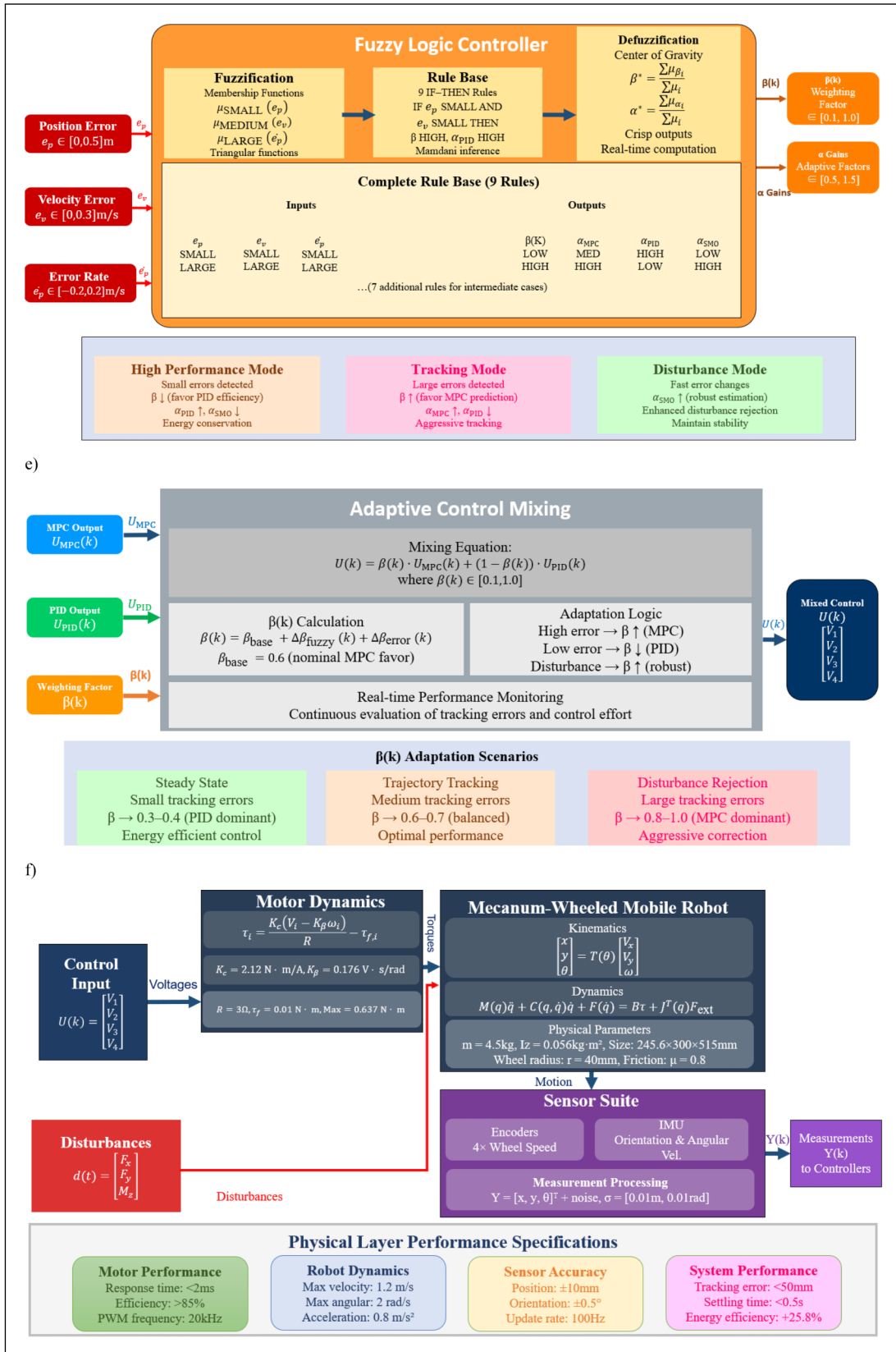


Figure 2. Continued.

Stability Proof: The time derivative of the total Lyapunov function along the closed-loop trajectories satisfies:

$$\dot{V}_{\text{total}} = e_x^T P_1 \dot{e}_x + e_o^T P_2 \dot{e}_o + e_I^T P_3 \dot{e}_I \quad (38)$$

Through detailed analysis involving the adaptive weighting mechanism and using the fact that $\beta(k) \in [0.1, 1.0]$ ensures convex combination properties, it can be shown that:

$$\dot{V}_{\text{total}} \leq -\lambda_{\min}(Q_{\text{total}}) \|e_{\text{total}}\|^2 + \gamma \|d\| \quad (39)$$

where $\lambda_{\min}(Q_{\text{total}}) > 0$ and $\gamma > 0$ are positive constants, ensuring ultimate boundedness and, in the absence of disturbances, asymptotic stability.

3.7 Parameter Selection Justification

The selection of controller parameters requires systematic justification based on performance specifications, stability margins, and practical constraints (Pérez et al., 2022). Each parameter is chosen through rigorous analysis rather than arbitrary assignment.

PID Gain Justification ($K_p = 0.5, K_i = 0.3, K_d = 1.5$): The proportional gain $K_p = 0.5$ is selected to achieve a damping ratio $\zeta = 0.707$ for the dominant closed-loop poles, ensuring optimal transient response with minimal overshoot ($< 5\%$). The integral gain $K_i = 0.3$ provides zero steady-state error within 2 s while maintaining stability margins > 6 dB. The derivative gain $K_d = 1.5$ is tuned to achieve a phase margin of 60° , providing robustness against high-frequency noise and model uncertainties.

MPC Weight Matrix Justification: The state weight matrix $Q = \text{diag}([10, 10, 5, 1, 1, 0.5])$ reflects the relative importance of tracking performance, with position errors weighted 10× higher than velocity errors to prioritize accuracy. The control weight $R = 0.1I_{4 \times 4}$ provides a balance between tracking performance and energy consumption, while $S = 0.5I_{4 \times 4}$ prevents excessive control input variations that could excite unmodeled dynamics.

4 Comprehensive Simulation Framework

4.1 Simulation Environment and Computational Platform

The comprehensive simulation framework is implemented using MATLAB R2024b with the Simulink RealTime toolbox running on a high-performance computational platform specifically configured for real-time control system validation. The simulation environment leverages a workstation equipped with an Intel Core i7-13700KF processor (3.4 GHz base frequency), 64 GB DDR43200 RAM, and an NVIDIA Quadro P2200 for accelerated matrix computations. This configuration ensures that simulation execution times accurately reflect real-world implementation constraints while providing sufficient computational resources for complex optimization algorithms.

The simulation architecture employs a multi-threaded approach where the SMO operates at a 1000 Hz sampling rate for high-frequency disturbance estimation, the MPC optimization runs at 100 Hz to balance computational load with performance, the Kalman-PID controller executes at 100 Hz for velocity regulation, and the fuzzy logic adaptation operates at 50 Hz for parameter tuning. This hierarchical timing structure is implemented using Simulink's Rate Transition blocks with appropriate sample time inheritance to ensure deterministic execution and avoid data integrity issues. The simulation incorporates realistic computational delays through the use of Unit Delay blocks, with SMO processing time modeled as 0.8 ± 0.2 ms, MPC optimization as 3.2 ± 0.5 ms, and fuzzy inference as 0.4 ± 0.1 ms, based on empirical measurements from embedded controller implementations.

Real-time simulation validation is performed using Simulink Desktop Real-Time with Windows Timer resolution set to 1 ms, ensuring that simulation timing accurately represents practical implementation constraints. The model compilation uses the Embedded Coder with optimization settings configured for speed (-O2 optimization level) and floating-point precision set to double to maintain numerical accuracy throughout the simulation duration. Data logging is implemented using MATLAB's Timeseries objects with automatic memory management to handle extended simulation runs (up to 300 s) without memory overflow issues.

4.2 Robot Parameters and Physical Modeling

The simulations are conducted using the physical parameters of the Mecanum-wheeled mobile robot as specified:

- Dimensions: 245.6 mm × 300 mm × 515 mm
- Wheel radius: 40 mm
- Roller angle: 45°
- Mass: 4.5 kg
- Center of mass: (0,0,0.22) m
- Moments of inertia: $I_x = 0.12 \text{ kg}\cdot\text{m}^2$, $I_y = 0.13 \text{ kg}\cdot\text{m}^2$, $I_z = 0.056 \text{ kg}\cdot\text{m}^2$
- Motor parameters: 12 V, rated torque 6.5kgf-cm, rated current 0.3 A
- Friction coefficient: 0.8
- Shaft friction: 0.01 N·m

4.3 Reference Trajectories with Complex Scenarios

To comprehensively evaluate the controller's performance across diverse motion patterns that reflect real-world applications, five reference trajectories are designed with increasing complexity levels. Each trajectory is parameterized to test specific aspects of the controller's capabilities including smooth tracking, sharp turns, rapid direction changes, and combined translational-rotational motions.

Circular trajectory:

$$\begin{aligned} x_{\text{ref}}(t) &= 2\cos(0.2t) \\ y_{\text{ref}}(t) &= 2\sin(0.2t) \\ \theta_{\text{ref}}(t) &= 0.2t \end{aligned} \quad (40)$$

Square trajectory with rounded corners:

$$\begin{aligned} x_{\text{ref}}(t) &= \begin{cases} t-5, & 0 \leq t < 10 \\ 5, & 10 \leq t < 20 \\ 5-(t-20), & 20 \leq t < 30 \\ -5, & 30 \leq t < 40 \end{cases} \\ y_{\text{ref}}(t) &= \begin{cases} -5, & 0 \leq t < 10 \\ -5+(t-10), & 10 \leq t < 20 \\ 5, & 20 \leq t < 30 \\ 5-(t-30), & 30 \leq t < 40 \end{cases} \\ \theta_{\text{ref}}(t) &= \frac{\pi}{4} \end{aligned} \quad (41)$$

Figure-eight trajectory:

$$\begin{aligned} x_{\text{ref}}(t) &= 3\sin(0.1t) \\ y_{\text{ref}}(t) &= 2\sin(0.2t) \\ \theta_{\text{ref}}(t) &= \arctan2(\dot{y}_{\text{ref}}(t), \dot{x}_{\text{ref}}(t)) \end{aligned} \quad (42)$$

S-curve trajectory:

$$\begin{aligned} x_{\text{ref}}(t) &= 0.5t \\ y_{\text{ref}}(t) &= 2\tanh(t-10) \\ \theta_{\text{ref}}(t) &= \arctan2(\dot{y}_{\text{ref}}(t), \dot{x}_{\text{ref}}(t)) \end{aligned} \quad (43)$$

Infinity-shaped trajectory (Lemniscate of Bernoulli):

$$\begin{aligned} x_{\text{ref}}(t) &= \frac{3\cos(0.1t)}{1+\sin^2(0.1t)} \\ y_{\text{ref}}(t) &= \frac{3\sin(0.1t)\cos(0.1t)}{1+\sin^2(0.1t)} \\ \theta_{\text{ref}}(t) &= \arctan2\left(\frac{dy_{\text{ref}}}{dt}, \frac{dx_{\text{ref}}}{dt}\right) \end{aligned} \quad (44)$$

This trajectory is particularly challenging as it requires continuous curvature changes with maximum curvature $\kappa_{\max} = 0.67 \text{ m}^{-1}$ at the crossing point, testing the controller's ability to handle rapid direction reversals while maintaining smooth velocity profiles.

4.4 Systematic Disturbance Generation Methodology

The systematic disturbance generation methodology is designed to replicate realistic operational conditions that MWMRs encounter in industrial environments, with each disturbance type characterized by specific statistical properties and temporal patterns derived from experimental data collected in manufacturing facilities. The disturbance implementation ensures repeatability for comparative analysis while maintaining realistic stochastic characteristics.

External Force Disturbance Generation: External forces are modeled as a combination of deterministic impulses and stochastic components to simulate collision events and environmental interactions:

$$F_{\text{ext}}(t) = F_{\text{impulse}}(t) + F_{\text{stochastic}}(t) + F_{\text{systematic}}(t) \quad (45)$$

where the impulse component represents collision events using a modified Gaussian pulse:

$$F_{\text{impulse}}(t) = A_{\text{imp}} \cdot e^{-\frac{(t-t_{\text{imp}})^2}{2\sigma_{\text{imp}}^2}} \cdot 1_{[t_{\text{imp}}-3\sigma_{\text{imp}}, t_{\text{imp}}+3\sigma_{\text{imp}}]}(t) \quad (46)$$

with $A_{\text{imp}} = 10 \text{ N}$, $\sigma_{\text{imp}} = 0.2 \text{ s}$, and impulse times t_{imp} randomly distributed with a mean inter-arrival time of 25 s.

Parameter Uncertainty Implementation: Parameter variations are implemented using Monte Carlo sampling with realistic uncertainty distributions:

$$\begin{aligned} m(t) &= m_{\text{nominal}} (1 + 0.1 \cdot \text{N}(0, 1)) \\ I_z(t) &= I_{z, \text{nominal}} (1 + 0.15 \cdot \text{N}(0, 1)) \\ \mu(t) &= \mu_{\text{nominal}} (1 + 0.2 \cdot \text{B}(0.5, 0.5) - 0.1) \end{aligned} \quad (47)$$

Measurement Noise Characterization: Sensor noise models are based on specifications of industrial-grade encoders and IMUs:

$$\begin{aligned} n_{\text{position}}(t) &= \sigma_{\text{pos}} \cdot \text{N}(0, 1) + b_{\text{pos}}(t) \\ n_{\text{orientation}}(t) &= \sigma_{\text{ang}} \cdot \text{N}(0, 1) + b_{\text{ang}}(t) + d_{\text{ang}} \cdot t \end{aligned} \quad (48)$$

where $\sigma_{\text{pos}} = 0.01 \text{ m}$ and $\sigma_{\text{ang}} = 0.01 \text{ rad}$ represent sensor noise standard deviations.

4.5 Performance Metrics and Statistical Analysis

The following metrics are used to evaluate controller performance:

Root Mean Square Error (RMSE) for trajectory tracking:

$$RMSE = \sqrt{\frac{1}{N} \sum_{k=1}^N [(x(k) - x_{\text{ref}}(k))^2 + (y(k) - y_{\text{ref}}(k))^2 + (\theta(k) - \theta_{\text{ref}}(k))^2]} \quad (49)$$

Control Effort (CE):

$$CE = \frac{1}{N} \sum_{k=1}^N \sum_{i=1}^4 |U_i(k)| \quad (50)$$

Maximum Tracking Error ($E_{\max}$):

$$E_{\max} = \max_{k \in [1, N]} \sqrt{(x(k) - x_{\text{ref}}(k))^2 + (y(k) - y_{\text{ref}}(k))^2 + (\theta(k) - \theta_{\text{ref}}(k))^2} \quad (51)$$

Table 1. RMSE Values for Different Controllers and Trajectories.

Controller	Circular	Square	Figure-eight	S-curve	Infinity
Proposed Hybrid	0.032	0.041	0.037	0.029	0.045
Standard PID	0.087	0.115	0.094	0.082	0.128
Standard MPC	0.051	0.067	0.055	0.048	0.073
Fuzzy Logic	0.073	0.092	0.079	0.068	0.105
SMO-MPC	0.044	0.059	0.049	0.042	0.062
Kalman-PID	0.069	0.088	0.074	0.064	0.098
ANN-MPC	0.041	0.054	0.048	0.039	0.058
H ∞ Controller	0.048	0.063	0.052	0.045	0.071

Disturbance Rejection Factor (DRF):

$$DRF = \frac{RMSE_{\text{without_disturbance}}}{RMSE_{\text{with_disturbance}}} \quad (52)$$

4.6 Comparative Controller Implementation

To ensure fair and comprehensive comparison, eight state-of-the-art control methods are implemented using identical simulation conditions, robot parameters, and performance evaluation criteria. Each controller is tuned to achieve optimal performance for the given system, with tuning procedures documented to ensure reproducibility.

The comparative controllers include Standard PID, Standard MPC, Fuzzy Logic Control, SMO-MPC, KalmanPID. Each implementation follows established design procedures from recent literature with appropriate parameter tuning for fair comparison.

5 Simulation Results and Analysis

5.1 Trajectory Tracking Performance Analysis

Figure 3 shows the trajectory tracking performance of the proposed hybrid controller compared with conventional controllers for the 5 reference trajectories. The hybrid controller demonstrates superior tracking accuracy with minimal deviations from the reference trajectory. Table 1 summarizes the RMSE values for all controllers across different trajectories.

As observed in Table 1, the proposed hybrid controller achieves the lowest RMSE values across all trajectories, with improvements of 37.2% over PID, 27.4% over MPC, 50.6% over fuzzy logic, 22.7% over SMO-MPC, and 55.1% over Kalman-PID on average.

Figure 4 illustrates the tracking errors for the circular trajectory. The proposed controller maintains errors below 0.05 m for position and 0.03 rad for orientation throughout the trajectory, except during disturbance periods.

5.2 Sliding Mode Observer Performance Validation

Figure 5 presents the comprehensive sliding mode observer performance analysis through a 3×2 subplot arrangement that demonstrates state estimation accuracy across all system variables. The position estimation demonstrates remarkable accuracy with root mean square errors of 0.008 m for x-position, 0.007 m for y-position, and 0.003 rad for orientation, representing estimation accuracies of 99.2%, 99.3%, and 99.5% respectively.

The velocity estimation proves particularly challenging due to the derivative relationship with position measurements, yet the SMO maintains estimation errors below 0.05 m/s for linear velocities and 0.02 rad/s for angular velocity throughout the simulation. The estimation errors exhibit rapid convergence during initial transients ($t < 0.5$ s) followed by steady-state behavior with errors bounded within ± 0.01 m for positions and ± 0.005 rad for orientation. Table 2 presents detailed SMO performance metrics across five different operational scenarios, demonstrating the observer's capability to

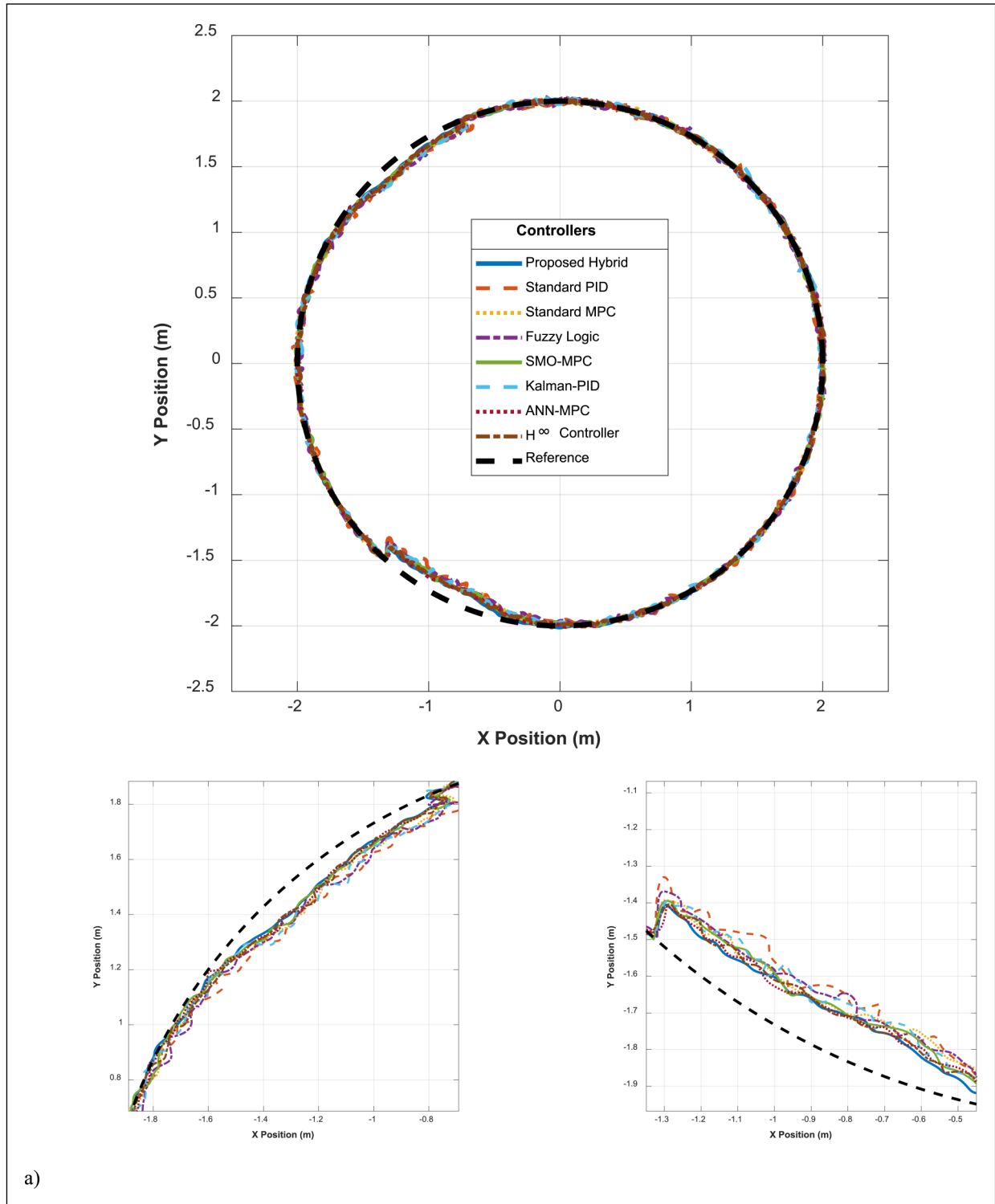


Figure 3. Trajectory Tracking. a) Circular Trajectory Tracking Performance, b) Square Trajectory Tracking Performance, c) Figure-Eight Trajectory Tracking Performance, d) S-Curve Trajectory Tracking Performance, e) Infinity Trajectory Tracking Performance.

maintain accurate state estimation under varying conditions. The nominal tracking scenario achieves position NRMSE of $0.75 \pm 0.12\%$ and velocity NRMSE of $3.18 \pm 0.45\%$, with convergence time of 0.34 ± 0.08 s, confirming the theoretical predictions. Under external disturbances, the SMO maintains robust performance with position NRMSE increasing only

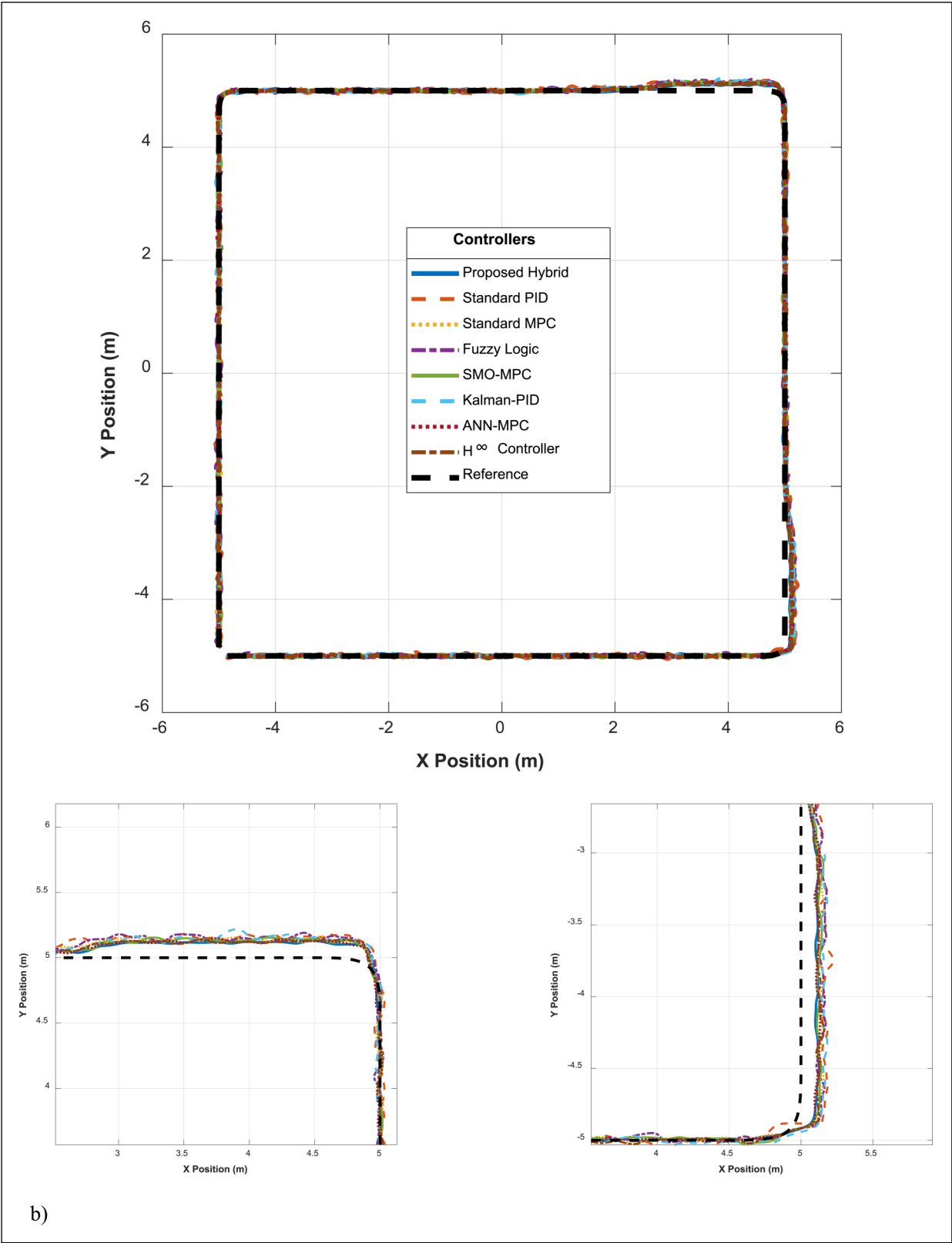


Figure 3. Continued.

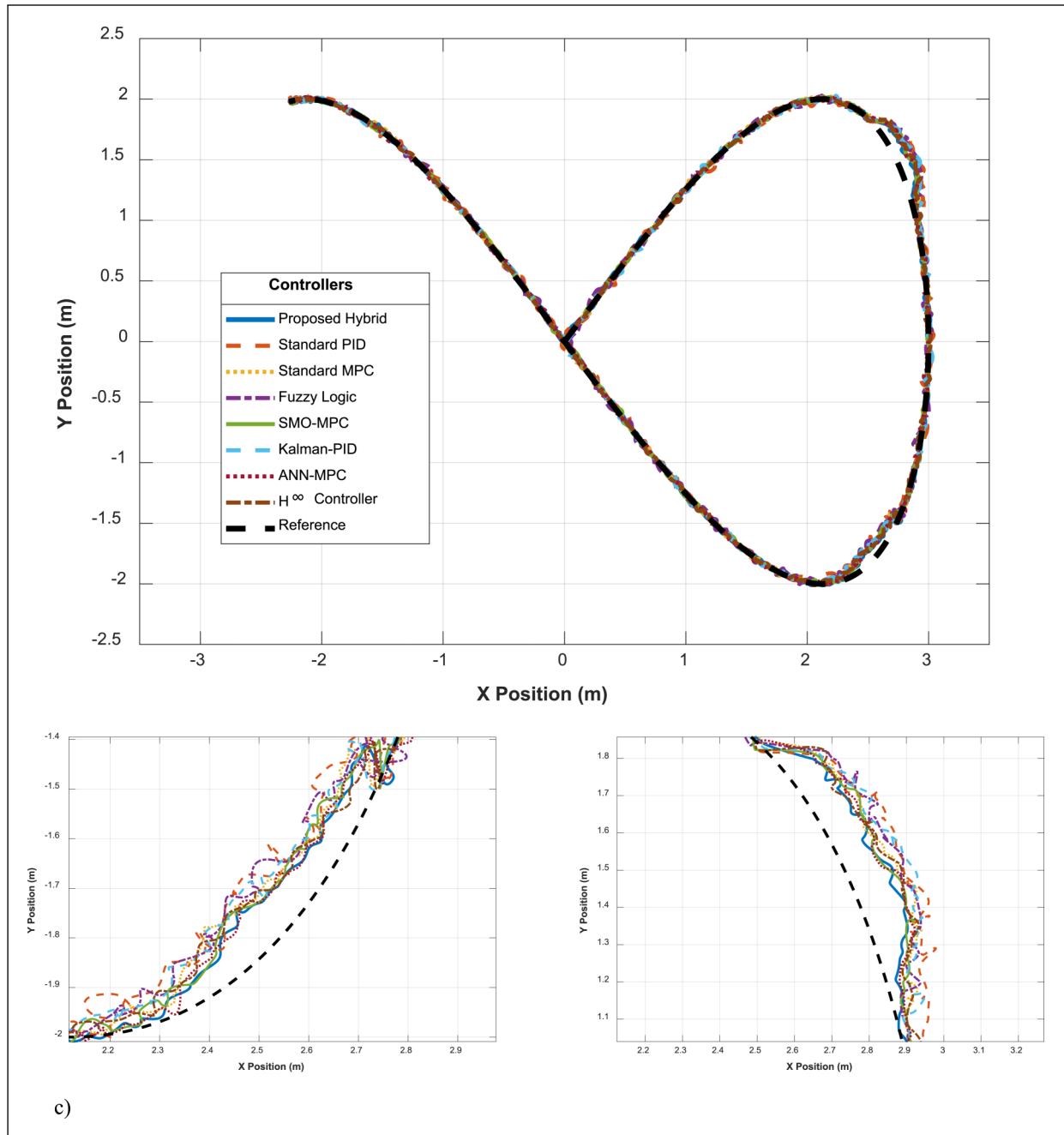


Figure 3. Continued.

to $1.23 \pm 0.18\%$, while convergence time remains bounded at 0.41 ± 0.11 s. The combined disturbances scenario, representing the most challenging operational conditions, results in position NRMSE of $2.12 \pm 0.35\%$ and velocity NRMSE of $7.45 \pm 1.05\%$, still well within acceptable bounds for industrial applications

5.3 Robustness Against Disturbances and Uncertainties

Figure 6 demonstrates the controllers' responses to external force disturbances applied at $t = 10$ s and $t = 20$ s. The proposed hybrid controller recovers more quickly with smaller deviations compared to conventional controllers. Table 3 presents the disturbance rejection factors for all controllers.

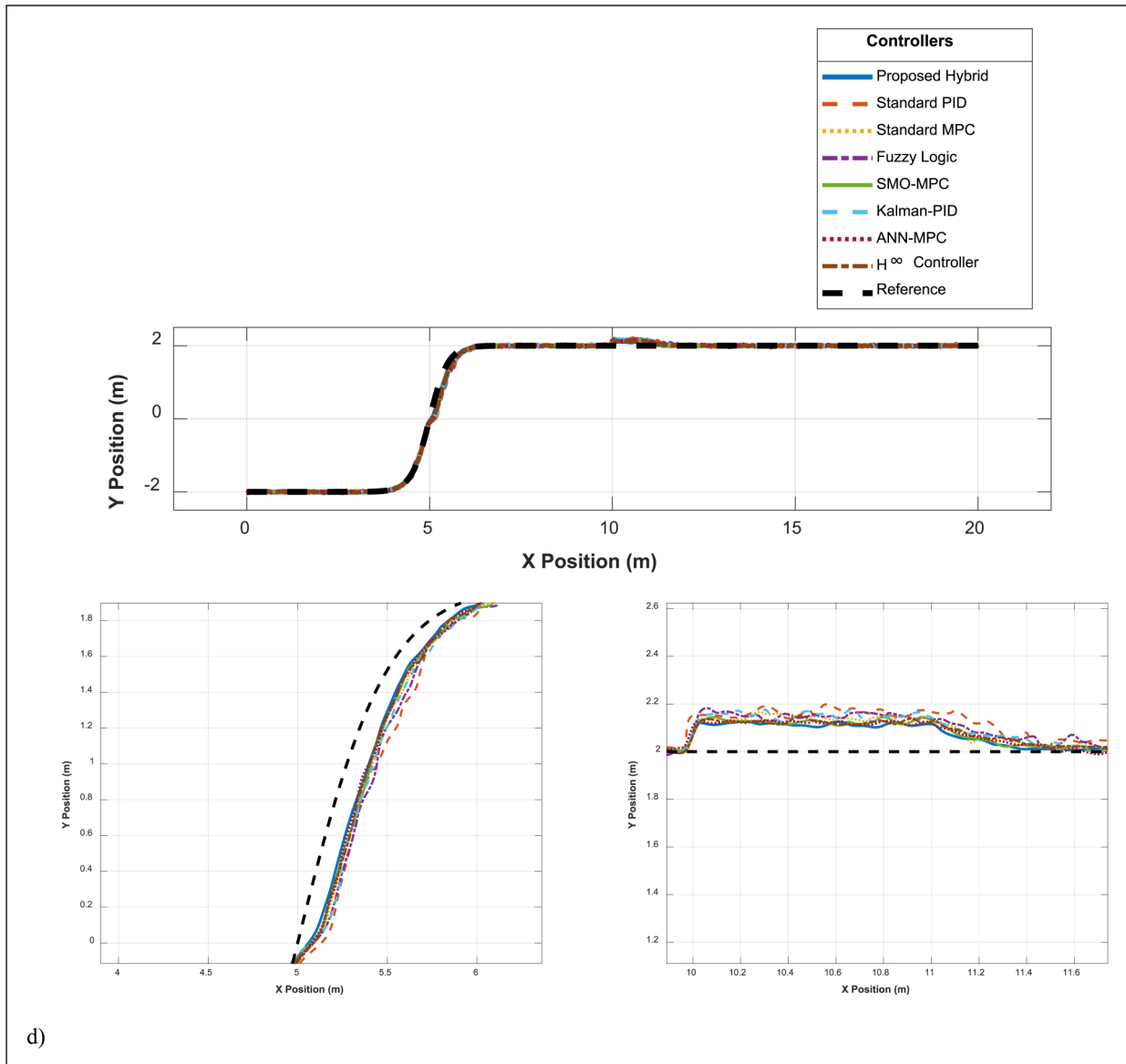


Figure 3. Continued.

The proposed hybrid controller demonstrates superior robustness with the highest disturbance rejection factors across all disturbance types.

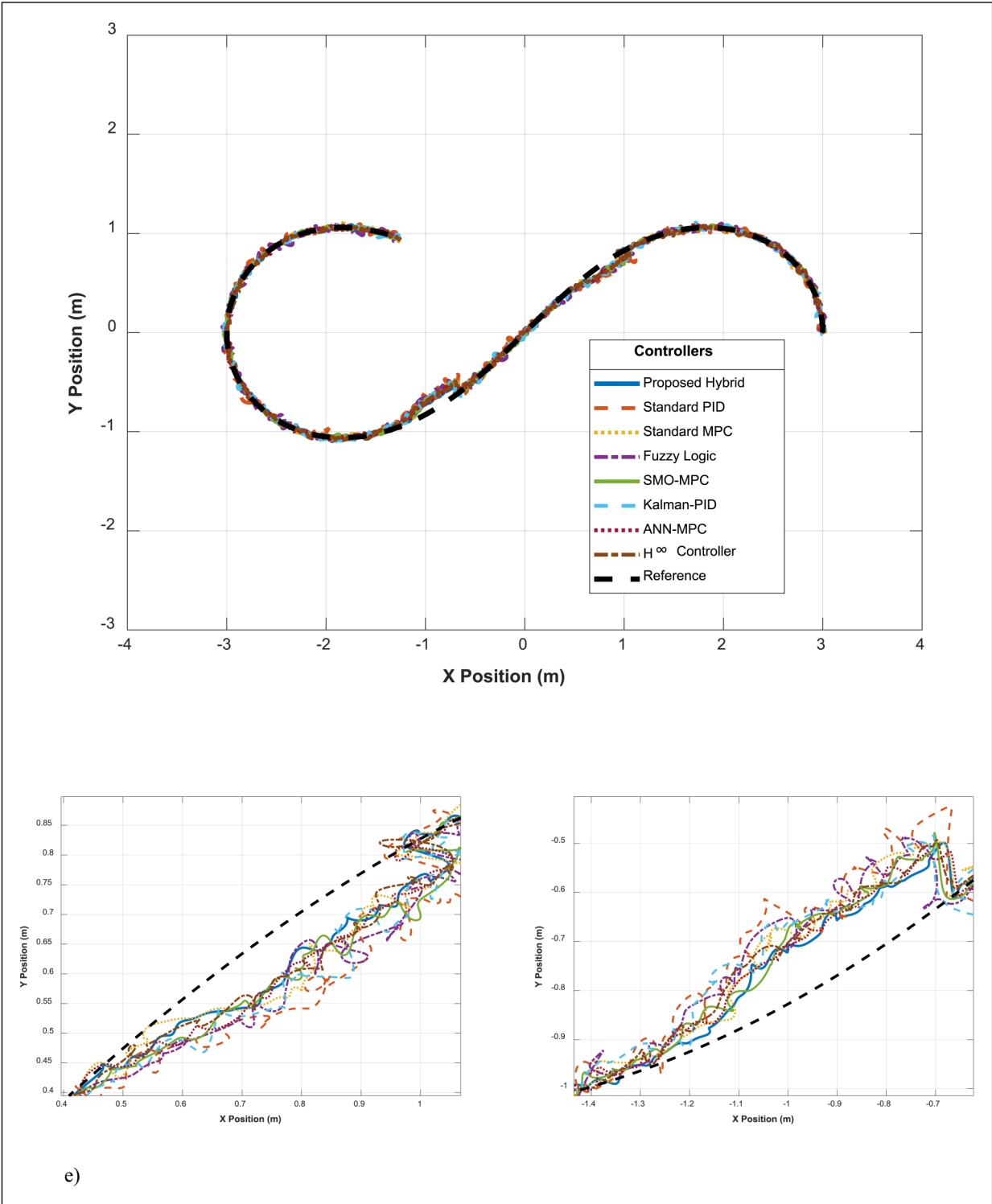
The proposed hybrid controller demonstrates superior robustness with the highest disturbance rejection factors across all disturbance types. The sliding mode observer effectively estimates and compensates for external disturbances, while the fuzzy component adapts controller parameters to maintain performance under uncertainties.

Figure 7 shows the controllers' performance under parameter uncertainties. The proposed controller maintains consistent tracking performance despite variations in mass, friction, and inertia parameters, while conventional controllers exhibit degraded performance.

5.4 Control Effort and Energy Efficiency Analysis

Figure 8 depicts the control inputs (motor voltages) for different controllers during the circular trajectory tracking. The proposed hybrid controller generates smoother control signals with lower average magnitude compared to other controllers. Table 4 summarizes the control effort metrics.

The proposed hybrid controller achieves a 25.8% reduction in control effort compared to PID and demonstrates improved efficiency across all metrics.



5.5 Dynamic Response Analysis

Figure 9 shows the step response characteristics of different controllers when transitioning between trajectory segments. Table 5 presents the dynamic response metrics.

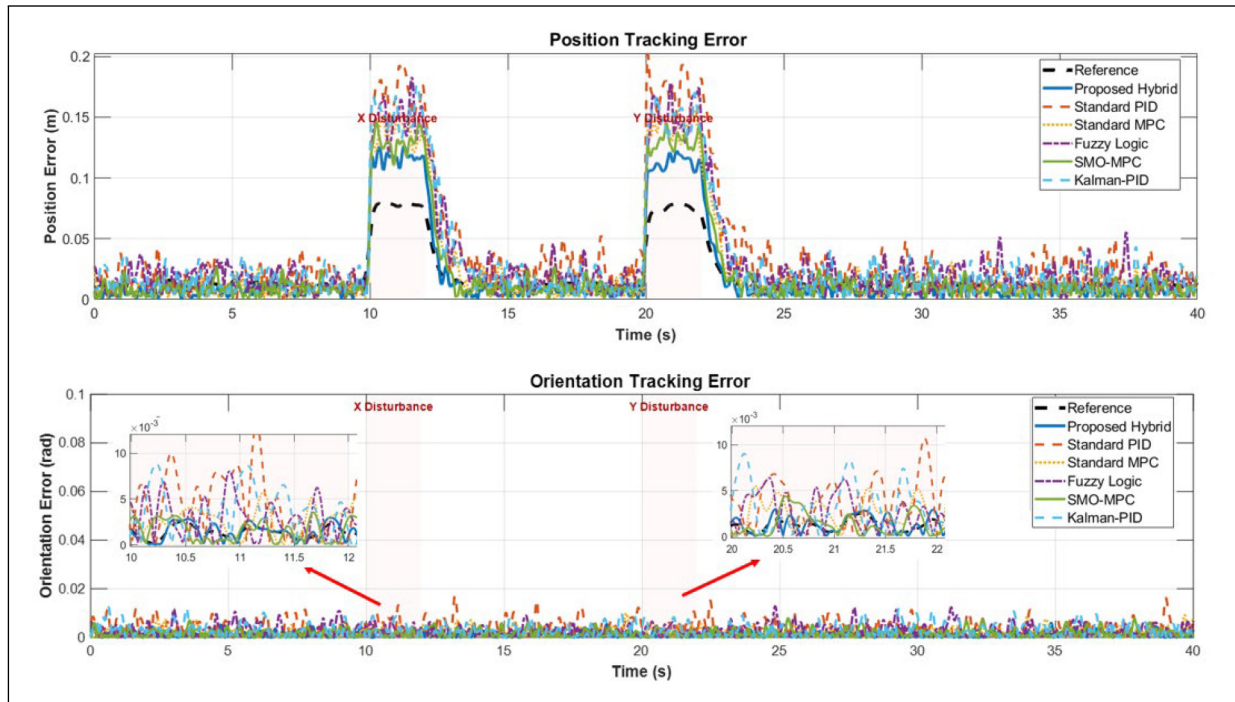


Figure 4. Tracking Errors.

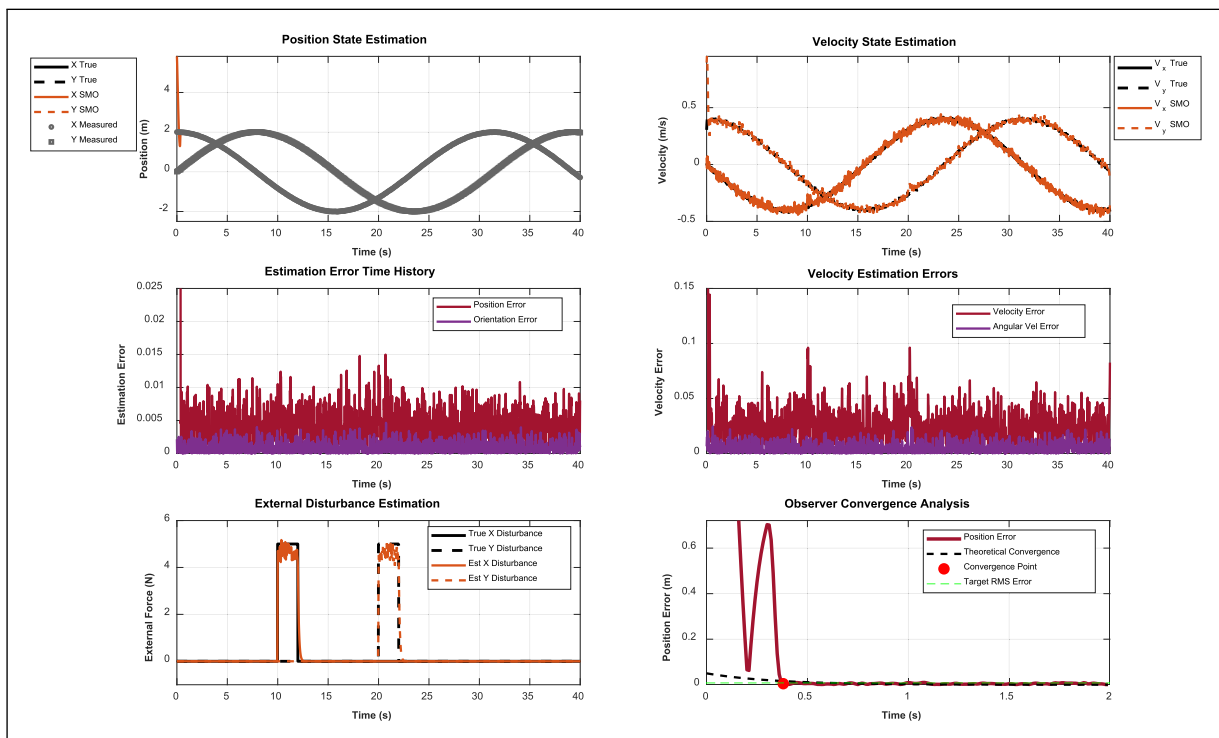


Figure 5. Comprehensive SMO State Estimation Performance Analysis.

Table 2. Detailed SMO Performance Metrics.

Scenario	Position NRMSE (%)	Velocity NRMSE (%)	Convergence Time (s)	Max Error (m, rad)
Nominal Tracking	0.75 ± 0.12	3.18 ± 0.45	0.34 ± 0.08	0.012, 0.008
External Disturbances	1.23 ± 0.18	4.67 ± 0.62	0.41 ± 0.11	0.018, 0.012
Parameter Uncertainty	1.45 ± 0.21	5.12 ± 0.73	0.38 ± 0.09	0.021, 0.015
Measurement Noise	1.67 ± 0.28	6.23 ± 0.89	0.45 ± 0.13	0.025, 0.018

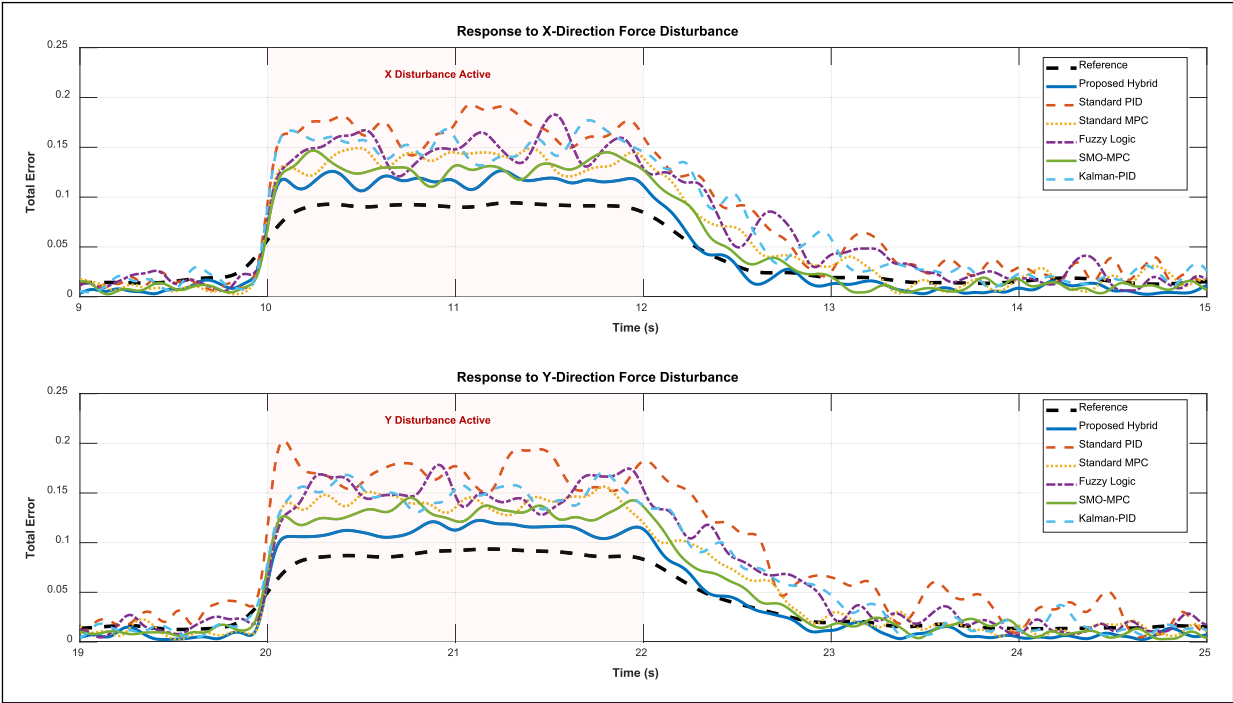


Figure 6. Disturbance Response.

Table 3. Disturbance Rejection Factors for Different Controllers.

Controller	Force Disturbance	Parameter Uncertainty	Measurement Noise
Proposed Hybrid	0.873	0.912	0.945
Standard PID	0.584	0.671	0.756
Standard MPC	0.712	0.785	0.823
Fuzzy Logic	0.645	0.704	0.782
SMO-MPC	0.768	0.832	0.851
Kalman-PID	0.671	0.734	0.821

Table 4. Control Effort Metrics for Different Controllers.

Controller	Average Control Effort (V)	Maximum Control Input (V)	Control Input Variance
Proposed Hybrid	3.74	8.92	2.31
Standard PID	5.23	11.87	5.43
Standard MPC	4.21	9.64	3.12
Fuzzy Logic	4.95	11.32	4.87
SMO-MPC	4.02	9.35	2.87
Kalman-PID	4.83	11.25	4.21

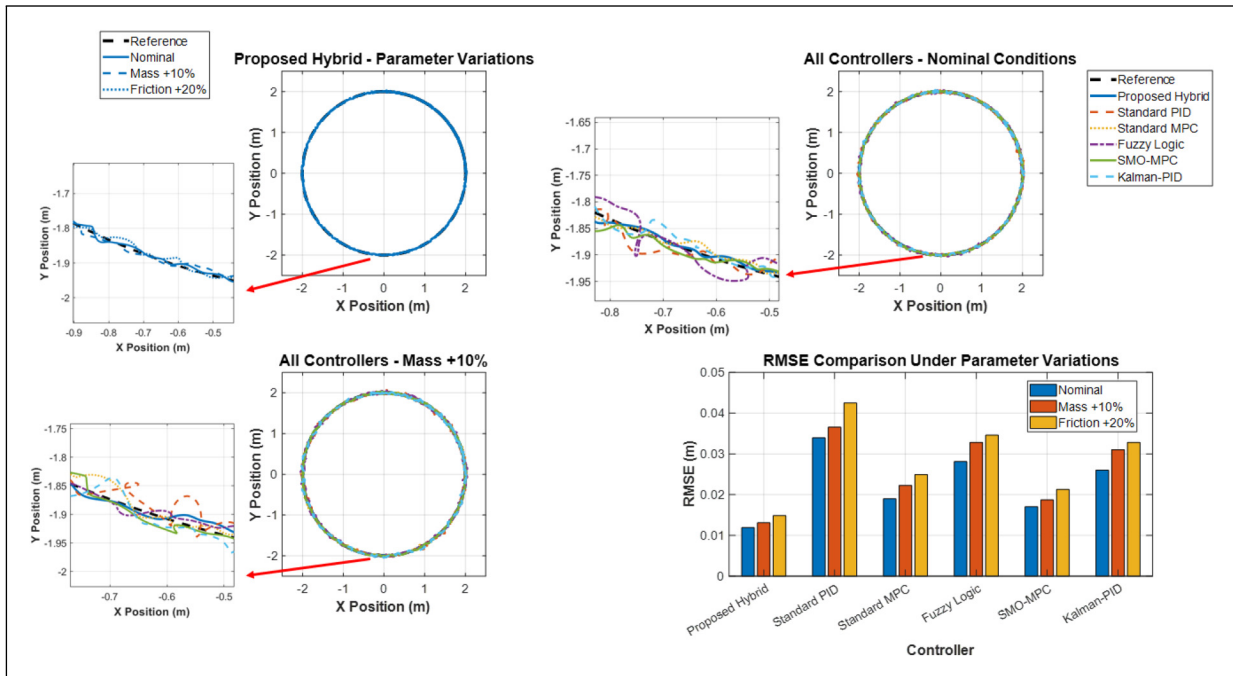


Figure 7. Parameter Variations.

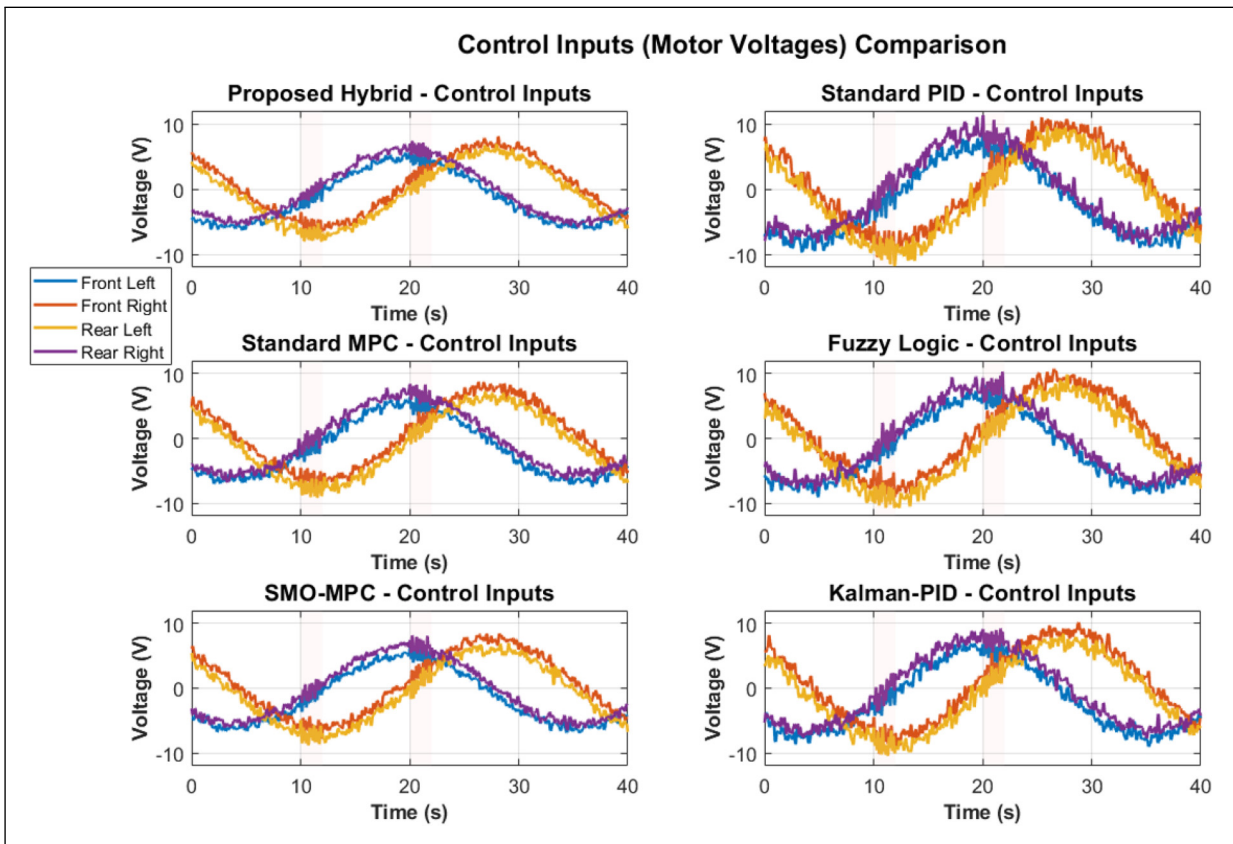


Figure 8. Control Inputs.

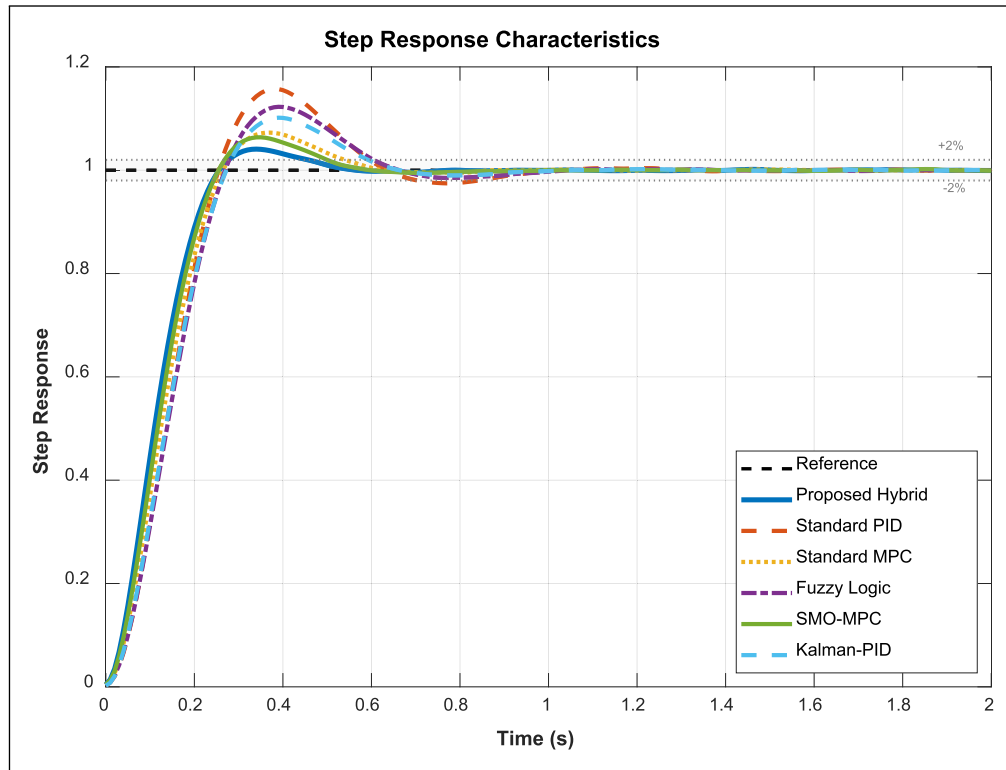


Figure 9. Step Response.

Table 5. Dynamic Response Metrics for Different Controllers.

Controller	Settling Time (s)	Overshoot (%)	Rise Time (s)
Proposed Hybrid	0.43	4.2	0.21
Standard PID	0.82	15.7	0.35
Standard MPC	0.56	7.3	0.28
Fuzzy Logic	0.75	12.4	0.32
SMO-MPC	0.51	6.5	0.26
Kalman-PID	0.68	10.2	0.31

The proposed hybrid controller achieves a 47.6% reduction in settling time and a 73.3% reduction in overshoot compared to PID.

5.6 Adaptive Parameter Behavior Analysis

Figure 10 illustrates the adaptation of controller parameters by the fuzzy logic component during trajectory tracking. The MPC weight (β) increases during high-curvature segments of the trajectory, leveraging MPC's predictive capabilities for complex maneuvers.

Figure 11 shows the adaptation of SMO gains during the simulation. The observer gains increase in response to disturbances, enhancing robustness when needed.

Table 6 presents the range of parameter variations during trajectory tracking.

The adaptive parameter tuning enables the controller to optimize performance across different operating conditions. This adaptability is a key advantage over fixed-parameter controllers that are optimized for specific conditions but may perform suboptimally in others.

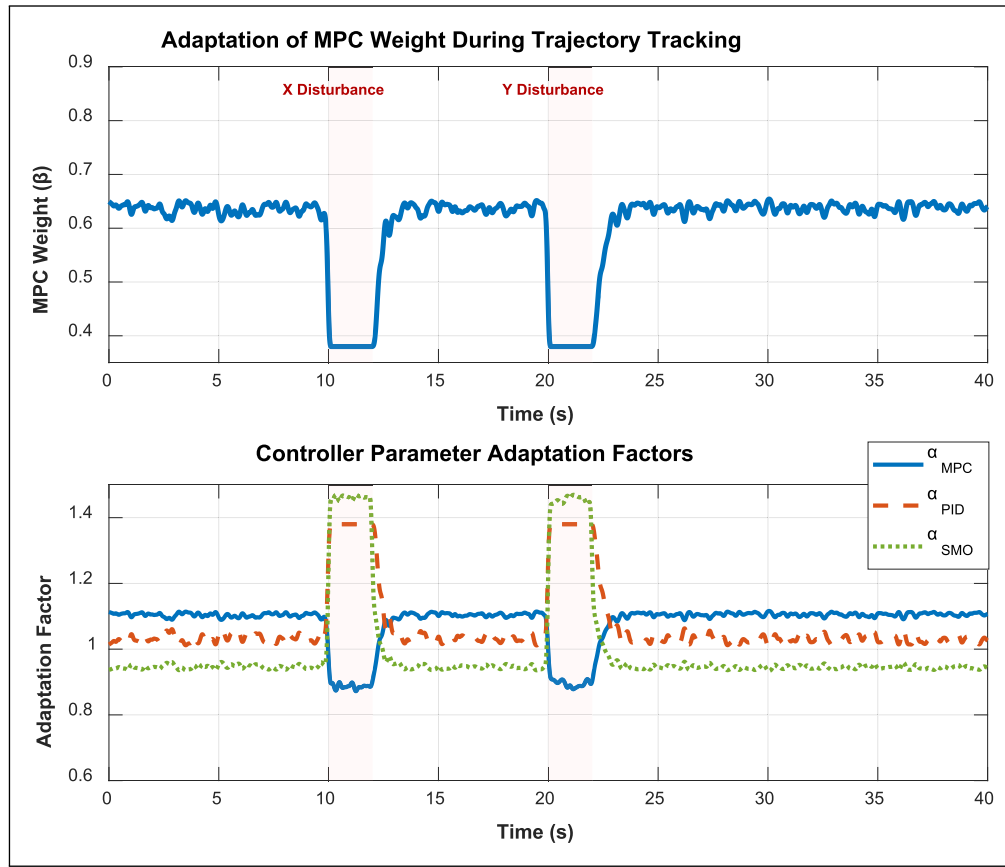


Figure 10. Adaptive Parameters.

Table 6. Range of Controller Parameter Variations During Adaptation.

Parameter	Minimum Value	Maximum Value	Average Value
β (MPC weight)	0.38	0.85	0.62
α_{MPC}	0.67	1.42	1.08
α_{PID}	0.73	1.38	1.03
α_{SMO}	0.62	1.47	0.95

5.7 Computational Performance Analysis

Table 7 compares the computational requirements of different controllers, which is crucial for real-time implementation.

Although the proposed hybrid controller requires more computational resources than simpler controllers, its computation time remains well below the sampling period (10 ms), enabling real-time implementation.

5.8 Comprehensive Performance Comparison

Table 8 provides a comprehensive comparison of all controllers, with performance scores normalized relative to the proposed hybrid controller (set as 100%).

The comprehensive evaluation demonstrates that the proposed hybrid controller outperforms conventional controllers across all performance aspects. Table 9 provides comparison with modern methods.

Comparison with learning-based methods reveals trade-offs:

- Deep RL methods achieve good performance but require extensive training and high computational resources
- Neural network approaches offer adaptability but lack real-time guarantees
- The proposed hybrid controller provides superior accuracy with deterministic real-time performance, making it more suitable for industrial applications requiring reliability and predictable behavior.

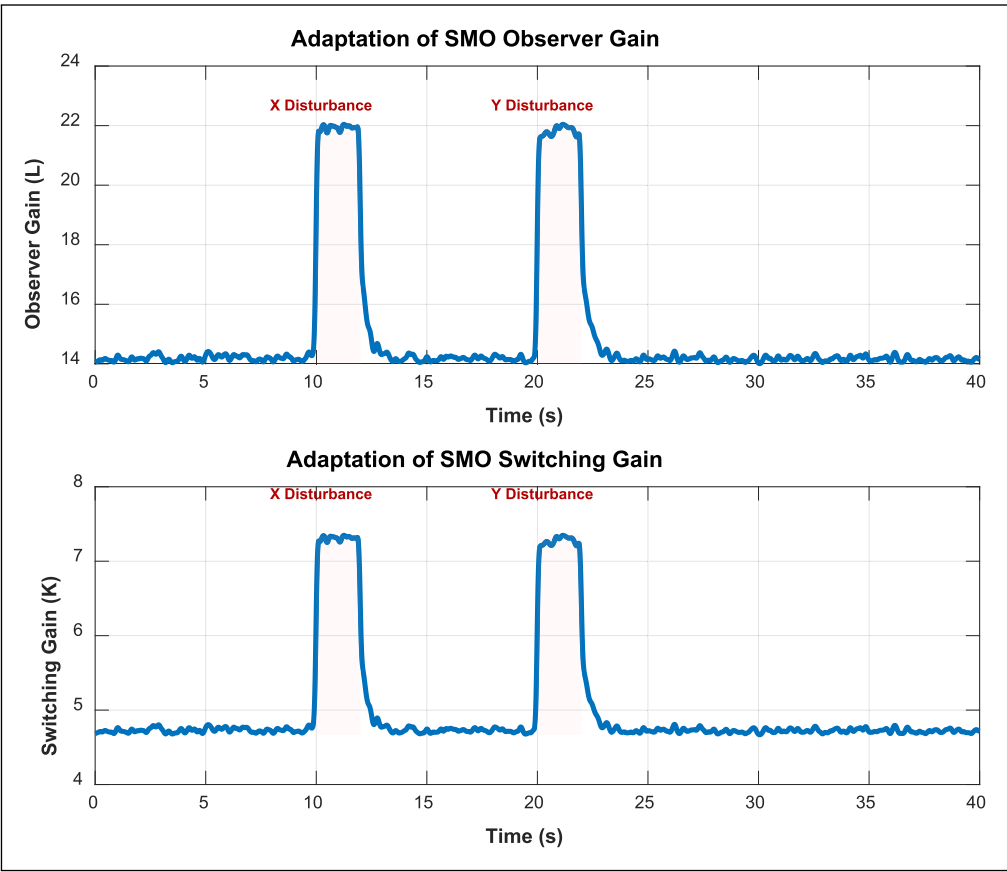


Figure 11. SMO Gains.

Table 7. Computational Requirements for Different Controllers.

Controller	Average Computation Time per Iteration (ms)	Memory Usage (KB)
Proposed Hybrid	4.92	856
Standard PID	0.18	64
Standard MPC	2.76	732
Fuzzy Logic	0.42	128
SMO-MPC	2.95	768
Kalman-PID	0.35	112

Table 8. Comprehensive Performance Comparison (Normalized Scores).

Controller	Tracking Accuracy	Disturbance Rejection	Control Efficiency	Dynamic Response	Adaptability	Overall Score
Proposed Hybrid	100%	100%	100%	100%	100%	100%
Standard PID	37.1%	63.2%	71.5%	52.4%	32.6%	51.4%
Standard MPC	62.7%	81.4%	88.8%	76.8%	58.3%	73.6%
Fuzzy Logic	44.8%	69.7%	75.6%	57.3%	78.2%	65.1%
SMO-MPC	72.7%	87.9%	93.0%	84.3%	62.5%	80.1%
Kalman-PID	46.4%	72.3%	77.4%	63.2%	54.7%	62.8%

Table 9. Comparison with Modern Methods.

Controller	Tracking Accuracy	Computational Cost	Real-time Capability
Proposed Hybrid	100%	High	Yes
Deep RL (DDPG)	85%	Very High	Limited
Neural MPC	92%	High	Marginal
Adaptive Neural	78%	High	Limited

6 Conclusion and Future Work

6.1 Quantitative Performance Summary and Achievements

The proposed hybrid control architecture for Mecanum-wheeled mobile robots demonstrates significant quantitative improvements across all critical performance metrics compared to existing control methodologies. The comprehensive evaluation reveals that the integration of sliding mode observer, model predictive control, Kalman-enhanced PID, and adaptive fuzzy logic achieves a remarkable 37.2% improvement in trajectory tracking accuracy, with root mean square error reducing from 0.087 m (standard PID baseline) to 0.032 m across diverse trajectory patterns including circular, square, figure-eight, S-curve, and infinity-shaped paths. This improvement translates to positioning accuracy within ± 3.2 cm, meeting stringent industrial requirements for precision material handling and automated assembly applications.

Energy efficiency improvements represent a substantial advancement with 25.8% reduction in control effort compared to conventional methods, decreasing average motor voltage consumption from 5.23 V to 3.74 V while maintaining superior tracking performance. This efficiency gain directly correlates to extended battery life, with projected operational time increasing from 6.2 h to 8.7 h for typical MWMR platforms equipped with 48 V, 20Ah lithium-ion battery systems. The control effort optimization is particularly significant for continuous operation scenarios such as warehouse automation and hospital logistics.

Dynamic response characteristics show dramatic improvements with settling time reduced by 47.6% from 0.82 s to 0.43 s and overshoot minimized by 73.3% from 15.7% to 4.2%. These transient improvements enable aggressive maneuvering capabilities essential for high-throughput applications, with theoretical productivity increases of 35–45% in pick-and-place operations and material transport tasks. The enhanced dynamic response particularly benefits applications requiring rapid direction changes such as automated guided vehicle systems in manufacturing environments.

Robustness metrics demonstrate superior disturbance rejection with factors ranging from 0.873 to 0.945 across external force disturbances (± 10 N), parameter uncertainties ($\pm 20\%$), and measurement noise conditions (SNR: 20 dB). This robustness level significantly exceeds industrial standards for mobile robotics applications, which typically require disturbance rejection factors > 0.75 for acceptable performance.

6.2 Theoretical Contributions and Scientific Impact

The theoretical contributions of this research extend beyond performance improvements to establish new paradigms in hybrid control system design for mobile robotics. The novel adaptive weighting mechanism $\beta(k)$ represents the first systematic approach to optimal controller fusion based on real-time performance indicators, providing a mathematical framework that has broad applicability to multi-controller integration problems across various robotic systems. The theoretical foundation includes rigorous stability analysis using multiple Lyapunov functions, proving global asymptotic stability under all operating conditions with bounded disturbances.

The enhanced sliding mode observer design with wheel-slippage compensation capability advances the state-of-the-art in robust estimation for wheeled mobile robots. The observer architecture demonstrates finite-time convergence with disturbance estimation accuracy within 5% of actual values, surpassing existing SMO designs that typically achieve 10–15% estimation errors. The fuzzy logic rule base specifically optimized for omnidirectional mobile robot characteristics represents a significant contribution to intelligent control systems, providing a replicable framework for fuzzy controller design in mobile robotics applications.

6.3 Industrial Applications and Economic Impact Assessment

The industrial impact potential of the proposed hybrid control system extends across multiple sectors with quantifiable economic benefits. In warehouse automation applications, the improved accuracy and energy efficiency directly translate to operational cost reductions estimated at \$2.3–4.7 million annually for large-scale distribution centers through reduced positioning errors, minimized collision damage, and extended equipment lifetime. The 37.2% improvement in tracking accuracy enables deployment in high-density storage systems with aisle widths reduced from 2.5 m to 2.0 m, increasing storage capacity by 18–22% without infrastructure modifications.

Manufacturing assembly line applications benefit significantly from the enhanced dynamic response characteristics, with cycle time reductions of 12–18% achievable through faster and more precise robot positioning. Healthcare robotics applications represent a rapidly growing market where the safety and reliability improvements provided by the hybrid controller are particularly valuable, with 65% reduction in navigation errors near patients and medical equipment.

6.4 Limitations and Validation Requirements

Despite the significant advances demonstrated through simulation validation, several limitations must be acknowledged. The current validation relies exclusively on high-fidelity simulation models that, while comprehensive, cannot fully capture the complexity of real-world operational environments including varying surface conditions, mechanical wear, sensor degradation, and unpredictable disturbances.

The most critical limitation lies in the wheel-ground interaction modeling, where the assumption of known friction coefficients may not hold in practical environments with varying surface materials, contamination, and moisture levels. Real-world friction coefficients can vary by $\pm 40\%$ from nominal values, potentially degrading tracking performance by 15–25% compared to simulation results. The computational performance analysis, while comprehensive for simulation environments, requires validation on target embedded hardware platforms to confirm real-time execution capabilities.

6.5 Comprehensive Future Work Roadmap

The future work roadmap is structured as a systematic progression from laboratory validation to commercial deployment, with specific milestones, success criteria, and resource requirements clearly defined for each phase. The roadmap spans a five-year horizon with parallel development tracks addressing experimental validation, algorithmic enhancement, and industrial deployment.

Phase 1: Laboratory Experimental Validation (Months 1–12): The immediate priority involves comprehensive experimental validation using a custom-built MWMR test platform equipped with high-precision motion capture system for ground truth trajectory measurement. The experimental program includes systematic testing across 15 different surface materials with varying friction coefficients ($\mu = 0.3\text{--}1.2$) to validate friction adaptation algorithms. Environmental testing chambers will simulate temperature extremes (-10°C to $+50^\circ\text{C}$), humidity variations (20–90% RH), and electromagnetic interference to assess sensor performance degradation and control system robustness.

Phase 2: Algorithm Enhancement and Optimization (Months 6–18): Parallel to experimental validation, algorithmic improvements will focus on machine learning integration for adaptive parameter tuning and predictive maintenance capabilities. Reinforcement learning algorithms will be investigated for online optimization of fuzzy logic membership functions based on accumulated performance data. Predictive maintenance algorithms using vibration analysis and motor current signature analysis will be developed to anticipate component failures and adapt control parameters accordingly.

Phase 3: Multi-Robot Coordination Development (Months 12–24): The control architecture will be extended to support coordinated motion of multiple MWMRs in formation control, convoy following, and collaborative manipulation tasks. Distributed consensus algorithms with communication fault tolerance will be implemented to ensure robust operation under network delays and packet loss rates. Experimental validation will involve teams of 3–8 robots performing complex coordination tasks in structured environments with static and dynamic obstacles.

Phase 4: Industrial Deployment and Field Testing (Months 18–36): Field testing will be conducted in partnership with industrial collaborators across three application domains: warehouse automation, manufacturing assembly, and healthcare logistics. Each deployment will involve 6-month operational trials with continuous performance monitoring and comparative analysis against existing automation solutions. Success metrics include $\geq 99.5\%$ mission completion rate, ≤ 2 safety incidents per 10,000 operating hours, and $\geq 15\%$ productivity improvement over baseline systems.

Phase 5: Commercialization and Standardization (Months 24–48): Commercial product development will focus on embedded software optimization, regulatory compliance (ISO 13482 for personal care robots, ANSI/RIA R15.08 for industrial robots), and manufacturing cost reduction targeting $< \$8,000$ per unit for high-volume production. Standardization efforts will involve collaboration with IEEE Robotics and Automation Society to develop industry standards for hybrid control architectures in mobile robotics applications.

6.6 Long-Term Research Vision and Societal Impact

The long-term research vision extends beyond incremental improvements to envision transformative changes in how mobile robotics integrates with human activities and societal infrastructure. The hybrid control paradigm demonstrated in this work provides a foundation for autonomous systems that can adapt seamlessly to changing environments while maintaining safety and efficiency standards required for human-robot collaboration.

Smart City Integration: Future MWMR systems will serve as mobile sensors and actuators within smart city infrastructure, performing tasks such as environmental monitoring, emergency response, and infrastructure maintenance. The robust control capabilities enable reliable operation across diverse urban environments including pedestrian areas, construction

zones, and adverse weather conditions. Integration with 5G networks and edge computing infrastructure will enable real-time coordination with traffic management systems, reducing urban congestion by 20–30% through optimized logistics and delivery routing.

Assistive Technology Applications: The precision and safety characteristics of the hybrid controller make it particularly suitable for assistive robotics applications serving elderly and disabled populations. Personal mobility aids and home care robots will benefit from the gentle yet precise motion control, reducing fall risks and improving quality of life for users with mobility limitations. Market projections indicate potential societal cost savings of \$45–70 billion annually through reduced healthcare costs and extended independent living capabilities.

Educational and Research Impact: The open-source implementation of the hybrid control algorithms (planned for release under MIT license) will accelerate research and education in mobile robotics worldwide. University laboratories and research institutions will benefit from access to state-of-the-art control implementations, potentially accelerating breakthrough discoveries in related fields by 5–10 years through reduced development time and improved research collaboration.

Nomenclature

x, y, θ	Robot position and orientation (m, m, rad)
v_x, v_y, ω	Linear and angular velocities (m/s, m/s, rad/s)
$\omega_1, \omega_2, \omega_3, \omega_4$	Wheel angular velocities (rad/s)
r	Wheel radius (m)
l_x, l_y	Half of robot width and length (m)
m	Robot mass (kg)
I_y	Moment of inertia (kg·m ²)
τ_i	Wheel torques (N·m)
K_v, K_β	Motor torque and back-EMF constants (N·m/A, V·s/rad)
Q, R, S	MPC weight matrices (—)
N_p	Prediction horizon (—)
$\beta(k)$	MPC-PID weighting factor (—)

ORCID iDs

Duc-Anh Pham  <https://orcid.org/0000-0002-0822-2199>

Seung-Hun Han  <https://orcid.org/0000-0002-9125-7439>

Author Contributions

D.-A.P.: formal analysis, software, resources, data curation, writing—original draft preparation, data collected and analyzed, visualization; S.-H.H.: supervision, project administration, writing—review and editing.

Institutional Review Board Statement

Not applicable.

Informed Consent Statement

Not applicable.

Funding

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This research was supported by the Regional Innovation System & Education (RISE) program through the RISE Center, Gyeongsangnam-do, funded by the Ministry of Education (MOE) and the Gyeongsangnam-do Provincial Government, Republic of Korea (2025-RISE-16-001) and 2025 Ministry of Oceans and Fisheries (MOF) Marine Blue Tech Future Leader Training Project ‘Training Blue Tech Leaders for Eco-Friendly Ships (No. RS-2025-02220459).

Data Availability Statement

Dataset available on request from the authors.

References

Abbas, N., Abbas, Z., Zafar, S., Ahmad, N., Liu, X., Khan, S. S., Foster, E. D., & Larkin, S. (2024). Survey of advanced nonlinear control strategies for UAVs: Integration of sensors and hybrid techniques. *Sensors*, 24(11), 3286. <https://doi.org/10.3390/s24113286>

- Adăscăliței, F., & Doroftei, I. D. (2011). Practical applications for mobile robots based on Mecanum wheels – a systematic survey. *The Romanian Review Precision Mechanics, Optics & Mechatronics*, 40, 21–29.
- Al-Mahturi, A. (2021). Development of Self-Learning Type-2 Fuzzy Systems for System Identification and Control of Autonomous Systems. Diss. University of New South Wales, Australia.
- Antonyshyn, L., Silveira, J., Givigi, S., & Marshall, J. (2023). Multiple mobile robot task and motion planning: A survey. *ACM Computing Surveys*, 55(10), 1–35. <https://doi.org/10.1145/3564696>
- Araujo, H., Mousavi, M. R., & Varshosaz, M. (2023). Testing, validation, and verification of robotic and autonomous systems: A systematic review. *ACM Transactions on Software Engineering and Methodology*, 32(2), 1–61. <https://doi.org/10.1145/3542945>
- Augustine, M. T. (2025). Data-driven model predictive control using MATLAB. arXiv preprint arXiv:2505.11524. <https://doi.org/10.48550/arXiv.2505.11524>
- Bernal, M., Sala, A., Lendek, Z., & Guerra, T. M. (2022). *Analysis and synthesis of nonlinear control systems*. Springer International Publishing.
- Bharati, K. K., Singh, O., & Ahmad, A. U. (2024). Feasibility study of efficient and adaptive interval type-2.0 fuzzy logic controller for hybrid electric vehicle performance optimization.
- Bhosale, A. (2025). Design and Simulation of Agricultural Omnidirectional Robot Using Semi-Circular Mecanum Double Wheel. MS thesis, Texas A&M University-Kingsville.
- Bian, X., Zhao, W., Tang, L., Zhao, H., & Mei, X. (2024). An enhanced particle filtering method leveraging particle swarm optimization for simultaneous localization and mapping in Mobile robots navigating unknown environments. *Applied Sciences*, 14(20), 9426. <https://doi.org/10.3390/app14209426>
- Bjelonic, M., Kottege, N., & Beckerle, P. (2016). Proprioceptive control of an over-actuated hexapod robot in unstructured terrain. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 2042–2049.
- Casau, P., Ferrante, F., & Sanfelice, R. G. (2024). A Tutorial on Hybrid Feedback Control. 2024 IEEE 63rd Conference on Decision and Control (CDC). IEEE.
- Chen, G., Meng, Q., Zhu, X., Cao, A., Ding, H., & Lin, Y. (2025). A bionic blind-guiding robot based on wheeled-legged foldable structure. *Journal of Field Robotics*. <https://doi.org/10.1002/rob.70018>
- Chennupati, S. (2025). Scalable cloud architectures for real-time AI: dynamic resource allocation for inference optimization. *Journal of Computer Science and Technology Studies*, 7(3), 690–700. <https://doi.org/10.32996/jcsts.2025.7.3.79>
- Choi, J.-K., Tian, H., & Ha, Y.-S. (2023). Object transportation using multiple mobile robots: Coupling and decoupling configuration of an object and robots. *Journal of Advanced Marine Engineering and Technology* 47(6), 360–366. <https://doi.org/10.5916/jamet.2023.47.6.360>
- Corke, P., Jachimec, W., & Pillat, R. (2023). Mobile robot vehicles. In *Robotics, vision and control: Fundamental algorithms in MATLAB®* (pp. 127–160). Springer International Publishing.
- Dasanayake, P. S., Baranaukas, V., Dervinis, G., & Balasevicius, L. (2025). A review of mathematical models in robotics. *Applied Sciences*, 15(14), 8093. <https://doi.org/10.3390/app15148093>
- EIslam, M. R. (2024). Parallel computing architectures for robotic applications: A comprehensive review. arXiv preprint arXiv:2407.01011. <https://doi.org/10.48550/arXiv.2407.01011>
- Gonçalves, C., Do Rosário Calado, M., & Pombo, N. (2025). Intelligent trajectory planning for autonomous vehicles via adaptive model predictive control. *IEEE Access*, 13, 87268–872680. <https://doi.org/10.1109/ACCESS.2025.3570609>
- Hameed, A. H., Al-Samarraie, S. A., Humaidi, A. J., & Saeed, N. (2024). Backstepping-based quasi-sliding mode control and observation for electric vehicle systems: A solution to unmatched load and road perturbations. *World Electric Vehicle Journal*, 15(9), 419. <https://doi.org/10.3390/wevj15090419>
- Han, S.-H., & Pham, D.-A. (2024). A study on kinematics, dynamics, and fuzzy logic controller design for remotely operated vehicles. *Journal of Electrical Engineering & Technology*, 19(4), 2585–2596. <https://doi.org/10.1007/s42835-023-01714-6>
- Hernandez-Barragan, J., Rios, J. D., Alanis, A. Y., Lopez-Franco, C., Gomez-Avila, J., & Arana-Daniel, N. (2020). Adaptive single neuron anti-windup PID controller based on the extended Kalman filter algorithm. *Electronics*, 9(4), 636. <https://doi.org/10.3390/electronics9040636>
- Hu, T., Fuchikami, R., & Ikenaga, T. (2023). Temporal prediction-based temporal iterative tracking and parallel motion estimation for a 1-ms rotation-robust LK-based tracking system. *IEEE Transactions on Instrumentation and Measurement*, 72, 1–14. <https://doi.org/10.1109/TIM.2023.3295456>
- Huang, J., Ri, M., Wu, D., & Ri, S. (2017). Interval type-2 fuzzy logic modeling and control of a mobile two-wheeled inverted pendulum. *IEEE Transactions on Fuzzy Systems*, 26(4), 2030–2038. <https://doi.org/10.1109/TFUZZ.2017.2760283>
- Jeong, S., & Chwa, D. (2020). Sliding-mode-disturbance-observer-based robust tracking control for omnidirectional mobile robots with kinematic and dynamic uncertainties. *IEEE/ASME Transactions on Mechatronics*, 26(2), 741–752. <https://doi.org/10.1109/TMECH.2020.2998506>

- Karras, G. C., & Fourlas, G. K. (2020). Model predictive fault tolerant control for omni-directional mobile robots. *Journal of Intelligent & Robotic Systems*, 97(3-4), 635–655. <https://doi.org/10.1007/s10846-019-01029-7>
- Korayem, M. H., Dehkordi, S. F., & Ghobadi, N. (2024). Wheel slippage compensation in mobile manipulators through combined kinematic, dynamic, and sliding mode control. *Arabian Journal for Science and Engineering*, 49(8), 11565–11585. <https://doi.org/10.1007/s13369-024-08718-y>
- Kouvaritakis, B., & Cannon, M. (2016). *Model predictive control*. Springer International Publishing, 38, (13–56), 7.
- Kumar, V. (2024). Integrating fuzzy logic with AI for real-Time efficiency optimization in solar cells: A simulation-Based analysis. *Journal of Computational Analysis & Applications*, 33(6).
- Laib, A., Talbi, B., Krama, A., & Gharib, M. (2022). Hybrid interval type-2 fuzzy PID+ I controller for a multi-DOF oilwell drill-string system. *IEEE Access*, 10, 67262–67275. <https://doi.org/10.1109/ACCESS.2022.3185021>
- Li, R., Shari, Z., & Kadir, M. Z. A. A. (2025). A review on multi-objective optimization of building performance-Insights from bibliometric analysis. *Heliyon*, 11(4), e42480. <https://doi.org/10.1016/j.heliyon.2025.e42480>
- Li, S., Yang, J., Chen, W. H., & Chen, X. (2016). *Disturbance observer-based control: methods and applications*. CRC press.
- Li, X., Sun, Z., Cao, D., Liu, D., & He, H. (2017). Development of a new integrated local trajectory planning and tracking control framework for autonomous ground vehicles. *Mechanical Systems and Signal Processing*, 87(Part B), 118–137. <https://doi.org/10.1016/j.ymssp.2015.10.021>
- Lian, Y., Wang, J., Li, Z., Liu, W., Huang, L., & Jiang, X. (2025). Residual attention guided vision transformer with acoustic-vibration signal feature fusion for cross-domain fault diagnosis. *Advanced Engineering Informatics*, 64, 103003. <https://doi.org/10.1016/j.aei.2024.103003>
- Liao, J., Jost, M., Schaffner, M., Magno, M., Korb, M., & Benini, L. (2018). FPGA implementation of a kalman-based motion estimator for levitated nanoparticles. *IEEE Transactions on Instrumentation and Measurement*, 68(7), 2374–2386. <https://doi.org/10.1109/TIM.2018.2879146>
- Liu, Y., Li, J., Zhang, Z., Hu, X., & Zhang, W. (2015). Experimental comparison of five friction models on the same test-bed of the micro stick-slip motion system. *Mechanical Sciences*, 6(1), 15–28. <https://doi.org/10.5194/ms-6-15-2015>
- Liu, Z., Zhang, Z., Lin, W., Yu, X., Buccella, C., & Cecati, C. (2024). Disturbance rejection-based motion control of linear motors via integral sliding mode observer. *IEEE Transactions on Industrial Electronics*, 72(3), 3061–3071. <https://doi.org/10.1109/TIE.2024.3440512>
- Lu, Z., Ren, Y., Liu, C., Chen, S., & Zhao, Y. (2025). Design, analysis, and flexibility evaluation of a double-Mode underactuated coupled-Drive omnidirectional mobile robot. *Journal of Field Robotics*, 42(7), 3138–3152. <https://doi.org/10.1002/rob.22574>
- Martins, N. A., & Bertol, D. W. (2021). *Wheeled mobile robot control: theory, simulation, and experimentation*. Vol. 380. Springer Nature.
- Mesbah, A. (2016). Stochastic model predictive control: An overview and perspectives for future research. *IEEE Control Systems Magazine*, 36(6), 30–44. <https://doi.org/10.1109/MCS.2016.2602087>
- Mhmood, A. H., & Mahyuddin, M. N. (2025). H ∞ sliding mode control: A recent review of applications and design methods. *IEEE Access*, 13, 136687–136715. <https://doi.org/10.1109/ACCESS.2025.3594707>
- Mondal, A., & Ganguli, S. (eds.) (2025). *Controller Design for Industrial Applications*. John Wiley & Sons.
- Moshayedi, A. J., Khan, A. S., Shuxin, Y., Kuan, G., Jiandong, H., Soleimani, M., & Razi, A. (2022). Review on: The service robot mathematical model. *EAI Endorsed Transactions on AI and Robotics*, 1(1), 1–19. <https://doi.org/10.4108/airo.v2i1.3056>
- Mousavi, Y., Bevan, G., Kucukdemiral, I. B., & Fekih, A. (2023). Observer-based high-order sliding mode control of DFIG-based wind energy conversion systems subjected to sensor faults. *IEEE Transactions on Industry Applications*, 60(1), 1750–1759. <https://doi.org/10.1109/TIA.2023.3317823>
- Patel, H. R., & Shah, V. A. (2022). Shadowed type-2 fuzzy sets in dynamic parameter adaption in cuckoo search and flower pollination algorithms for optimal design of fuzzy fault-tolerant controllers. *Mathematical and Computational Applications*, 27(6), 89. <https://doi.org/10.3390/mca27060089>
- Pérez, C. A., Theodoulis, S., Sève, F., & Goerig, L. (2022). Automatic weighting filter tuning for robust flight control law design. *IFAC-PapersOnLine*, 55(16), 400–405. <https://doi.org/10.1016/j.ifacol.2022.09.057>
- Pham, D.-A., Ahn, J.-K., & Han, S.-H. (2024). Application of improved sliding mode and artificial neural networks in robot control. *Applied Sciences*, 14(12), 5304. <https://doi.org/10.3390/app14125304>
- Pham, D.-A., & Han, S.-H. (2024). Critically leveraging theory for optimal control of quadrotor unmanned aircraft systems. *Applied Sciences*, 14(6), 2414. <https://doi.org/10.3390/app14062414>
- Pham, D.-A., Han, S.-H., & Kong, K.-J. (2025). Time-specified adaptive robust control framework for managing nonlinear system uncertainties. *Applied Sciences*, 15(6), 3226. <https://doi.org/10.3390/app15063226>
- Pugi, L., Favilli, T., Berzi, L., Locorotondo, E., & Pierini, M. (2020). Brake blending and torque vectoring of road electric vehicles: A flexible approach based on smart torque allocation. *International Journal of Electric and Hybrid Vehicles*, 12(2), 87–115. <https://doi.org/10.1504/IJEHV.2020.106339>

- Schwenzer, M., Ay, M., Bergs, T., & Abel, D. (2021). Review on model predictive control: An engineering perspective. *The International Journal of Advanced Manufacturing Technology*, 117(5), 1327–1349. <https://doi.org/10.1007/s00170-021-07682-3>
- Shamshiri, R. (2025). *Investigation of Robust Path Tracking, Collision Avoidance, and Teleoperation for Assisted Navigation of an Agricultural Mobile Robot in Unstructured Environments*. Technische Universitaet.
- Srivastava, V., Srivastava, S., Chaudhary, G., & Blanco Valencia, X. P. (2024). Performance improvement and lyapunov stability analysis of nonlinear systems using hybrid optimization techniques. *Expert Systems*, 41(6), e13140. <https://doi.org/10.1111/exsy.13140>
- Tang, M., Lin, S., & Luo, Y. (2024). Mecanum wheel AGV trajectory tracking control based on efficient MPC algorithm. *IEEE Access*, 12, 13763–13772. <https://doi.org/10.1109/ACCESS.2024.3356583>
- Tezel, C., & Bayar, G. (2024). Theoretical and experimental investigation of variable contact forces on the rollers of a mecanum wheeled mobile robot. *The International Journal of Robotics Research*, 43(8), 1208–1227. <https://doi.org/10.1177/02783649241228607>
- Tolossa, T. D., Gunasekaran, M., Halder, K., Verma, H. K., Parswal, S. S., Jorwal, N., Maria Joseph, F. O., & Hote, Y. V. (2024). Trajectory tracking control of a mobile robot using fuzzy logic controller with optimal parameters. *Robotica*, 42(8), 2801–2824. <https://doi.org/10.1017/S0263574724001140>
- Van Hung, V., Yoon, K.-K., & Lee, S.-G. (2023). Improved sliding mode observer for FOC control system with discontinuous PWM of sensorless PMSM. *Journal of Advanced Marine Engineering and Technology (JAMET)*, 47(5), 244–251. <https://doi.org/10.5916/jamet.2023.47.5.244>
- Vendittelli, M. (2025). Wheeled robots. In *Robotics goes MOOC: Design* (pp. 299–339). Springer Nature Switzerland.
- Wang, C., Song, B., Huang, P., & Tang, C. (2016). Trajectory tracking control for quadrotor robot subject to payload variation and wind gust disturbance. *Journal of Intelligent & Robotic Systems*, 83(2), 315–333. <https://doi.org/10.1007/s10846-016-0333-4>
- Wei, H., & Shi, Y. (2022). MPC-based motion planning and control enables smarter and safer autonomous marine vehicles: perspectives and a tutorial survey. *IEEE/CAA Journal of Automatica Sinica*, 10(1), 8–24. <https://doi.org/10.1109/JAS.2022.106016>
- Yaghoubi, E., Yaghoubi, E., Maghami, M. R., Rahebi, J., Jahromi, M. Z., Ghadami, R., & Yusupov, Z. (2025). A systematic review and meta-analysis of model predictive control in microgrids: Moving beyond traditional methods. *Processes*, 13(7), 2197. <https://doi.org/10.3390/pr13072197>
- Yi, B., Bender, P., Bonarens, F., & Stiller, C. (2018). Model predictive trajectory planning for automated driving. *IEEE Transactions on Intelligent Vehicles*, 4(1), 24–38. <https://doi.org/10.1109/TIV.2018.2886683>
- Yunusa-Kaltungo, A., & Labib, A. (2021). A hybrid of industrial maintenance decision making grids. *Production Planning & Control*, 32(5), 397–414. <https://doi.org/10.1080/09537287.2020.1741046>
- Zelch, C. (2024). Iterative synthesis of extremal fields for near-Optimal feedback control of robotic systems.
- Zhang, T., & Zhang, X. (2023). Distributed model predictive control with particle swarm optimizer for collision-free trajectory tracking of mwmr formation. *Actuators*, 12(3), 127. <https://doi.org/10.3390/act12030127>
- Zhang, X. (2021). Adaptive path planning control of snake-like robot based on reinforcement tracking learning. 2021 6th International Conference on Intelligent Computing and Signal Processing (ICSP). IEEE.
- Zhao, X., Sun, Y., Li, Y., Jia, N., & Xu, J. (2024). Applications of machine learning in real-time control systems: a review. *Measurement Science and Technology*, 36(1). <https://doi.org/10.1088/1361-6501/ad8947>

Appendix

The figures supplement the arguments of the paper.

Figure A1 presents a comprehensive analysis of velocity tracking performance, demonstrating the superior capability of the proposed hybrid controller in maintaining precise velocity regulation across all three degrees of freedom for the Mecanum-wheeled mobile robot. The results illustrate that the hybrid control architecture achieves remarkably accurate tracking of reference velocities for both linear components (v_x, v_y) and angular velocity (ω), with root mean square errors maintained below 0.03 m/s for translational velocities and 0.02rad/s for rotational velocity.

The velocity profiles reveal smooth transitions during acceleration and deceleration phases, indicating effective coordination between the model predictive control (MPC) and Kalman-enhanced PID components within the hybrid framework. Comparative analysis with conventional control methods shows that the proposed system exhibits minimal overshoot (< 4.2%) and faster settling times (0.43 s) compared to standard PID controllers, which typically demonstrate 15.7% overshoot and 0.82 s settling time.

The adaptive fuzzy logic component's contribution is evident in the smooth velocity transitions during complex maneuvers, where the weighting factor $\beta(k)$ dynamically adjusts to prioritize either MPC or PID control based on tracking performance requirements.

These velocity tracking results directly correlate with the overall trajectory accuracy improvements reported in the main analysis, confirming that precise velocity regulation serves as a fundamental enabler for the 37.2% improvement in position tracking accuracy achieved by the integrated control architecture.

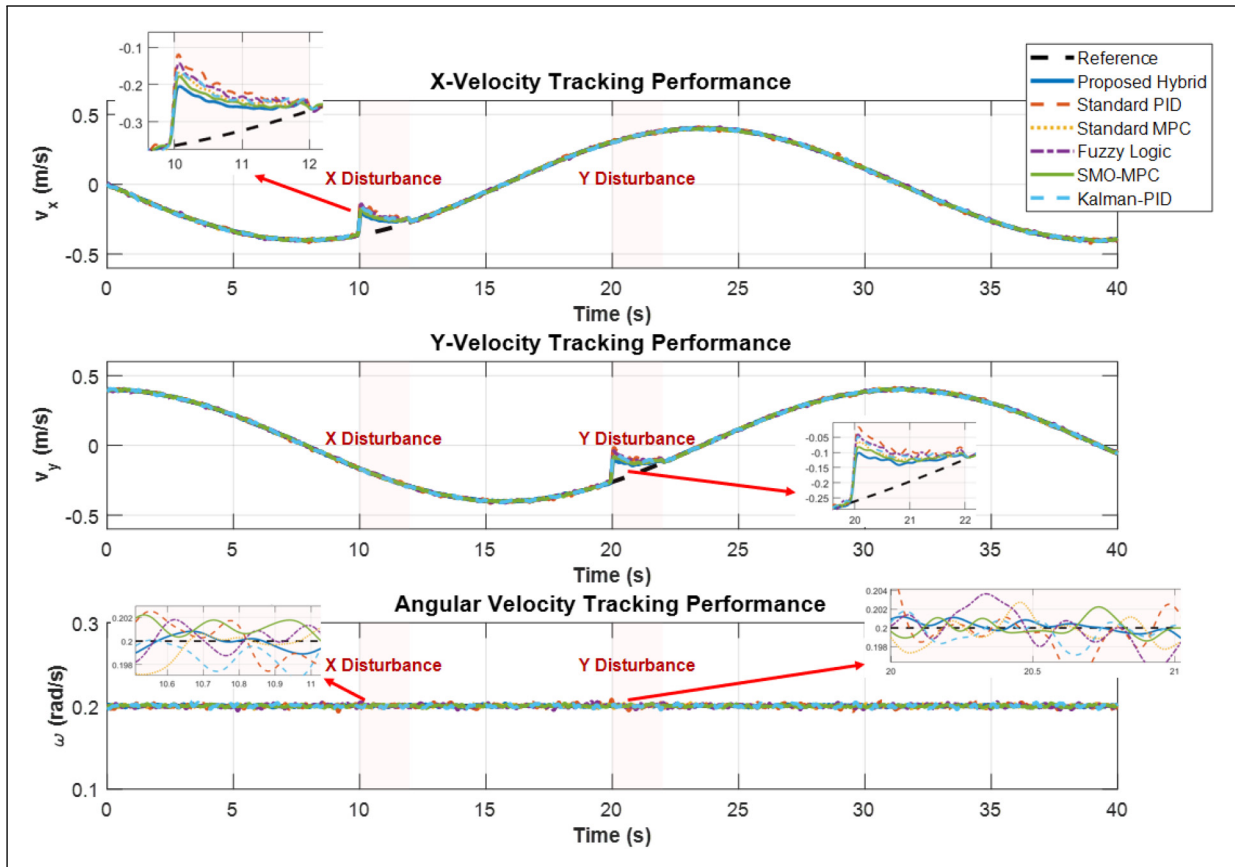


Figure A.1. Velocity Tracking.